Cranial Nerves and Brainstem - I

Aftab Merchant, MBBS, FCPS, MHPE
Professor of Medical Education
Assistant Dean of Anatomical Education
Kirk Kerkorian School of Medicine
University of Nevada Las Vegas

Learning Objectives

1. Describe the external anatomy of the brainstem, including the midbrain, pons, medulla, major surface landmarks, cranial nerve attachment sites, and major arterial supply. (CLO 1)
2. Describe the anatomical pathways, functional components, and major functions of the twelve cranial nerves, including their passage through skull foramina and clinically relevant head and neck structures, and correlate peripheral cranial nerve lesions with their characteristic neurological deficits. (CLO 1, CLO 2, CLO 3)
3. Explain the internal organization of the brainstem, including the arrangement of cranial nerve nuclei, cranial nerve fascicles, major ascending and descending pathways, and vascular territories, and describe their roles in normal sensory, motor, and autonomic function. (CLO 1, CLO 2)

Pre-Session Assignment

- Preview the PowerPoint slides. You are not expected to memorize details before class.
- B&B: Brainstem (30 min)
- B&B: Cranial Nerves I (21 min)
- B&B: Cranial Nerves II (21 min)

Brainstem Framework

External Anatomy • Cranial Nerve Attachments • Internal Organization • Blood Supply

Brainstem Organization

Three Divisions

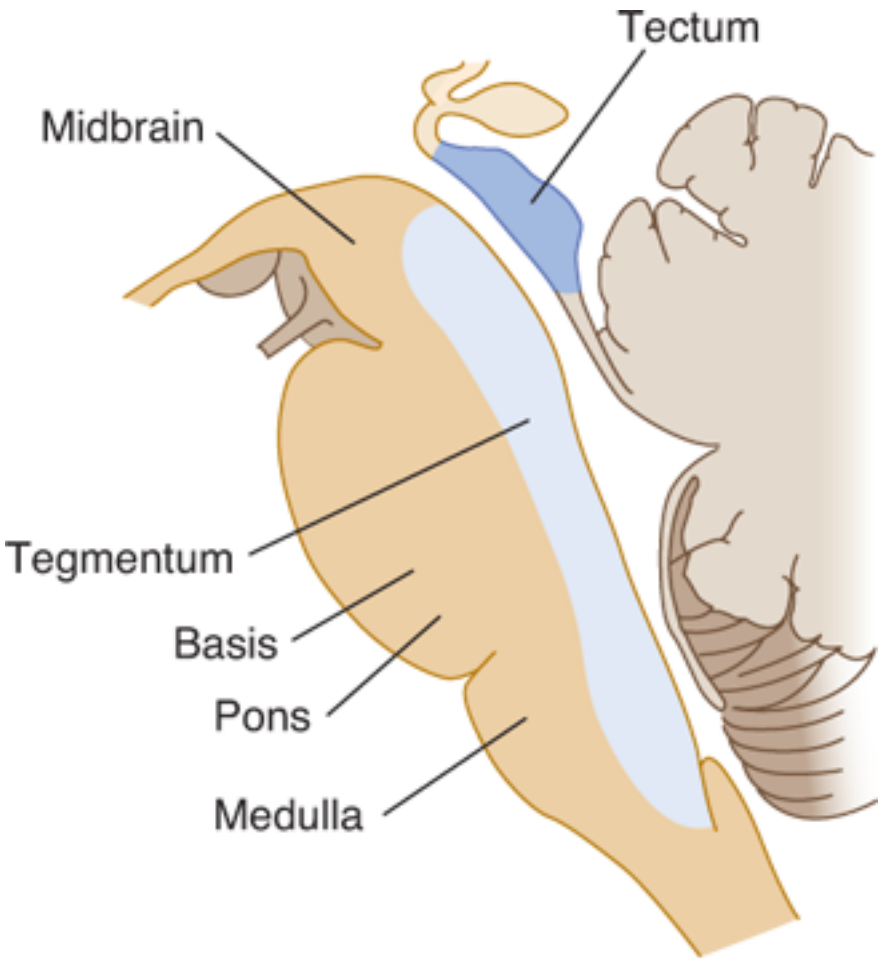
- Medulla oblongata
- Pons
- Midbrain

Internal Organization

- Tectum: dorsal midbrain
- Tegmentum: central
- Basis: ventral

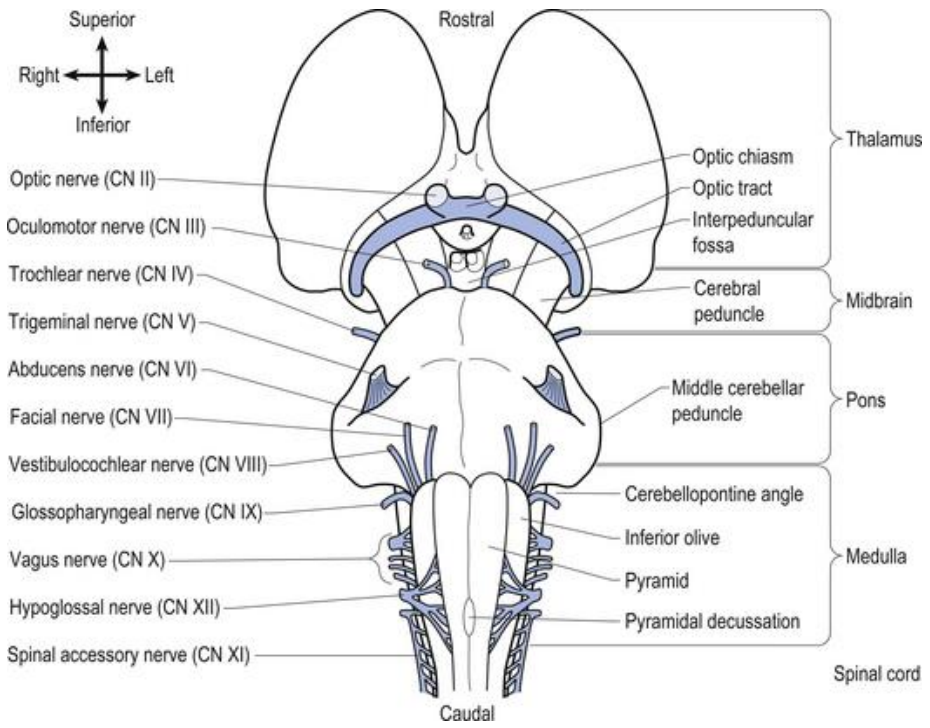
First orient yourself:

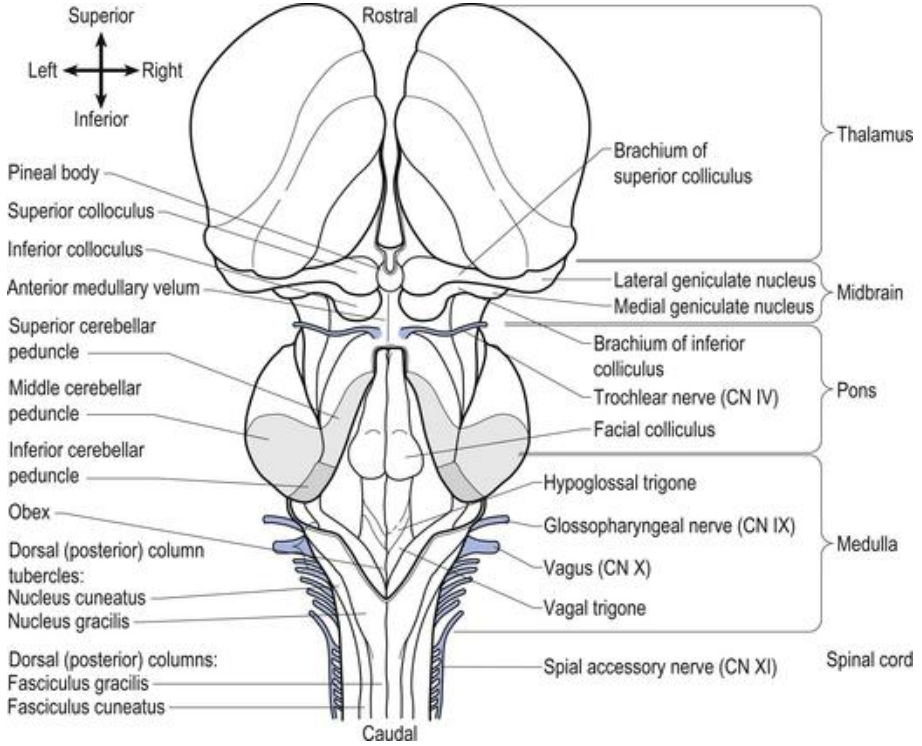
level → dorsal/ventral → medial/lateral



Source: Stephen G. Waxman
Clinical Neuroanatomy, Twenty-Eighth Edition
www.accessmedicine.com
Copyright © McGraw-Hill Education. All rights reserved.

External Brainstem Anatomy





Midbrain

- Cerebral peduncles
- Superior and inferior colliculi

Pons

- Basilar pons
- Middle cerebellar peduncles

Medulla

- Pyramids
- Olives

Cranial Nerve Attachment Sites

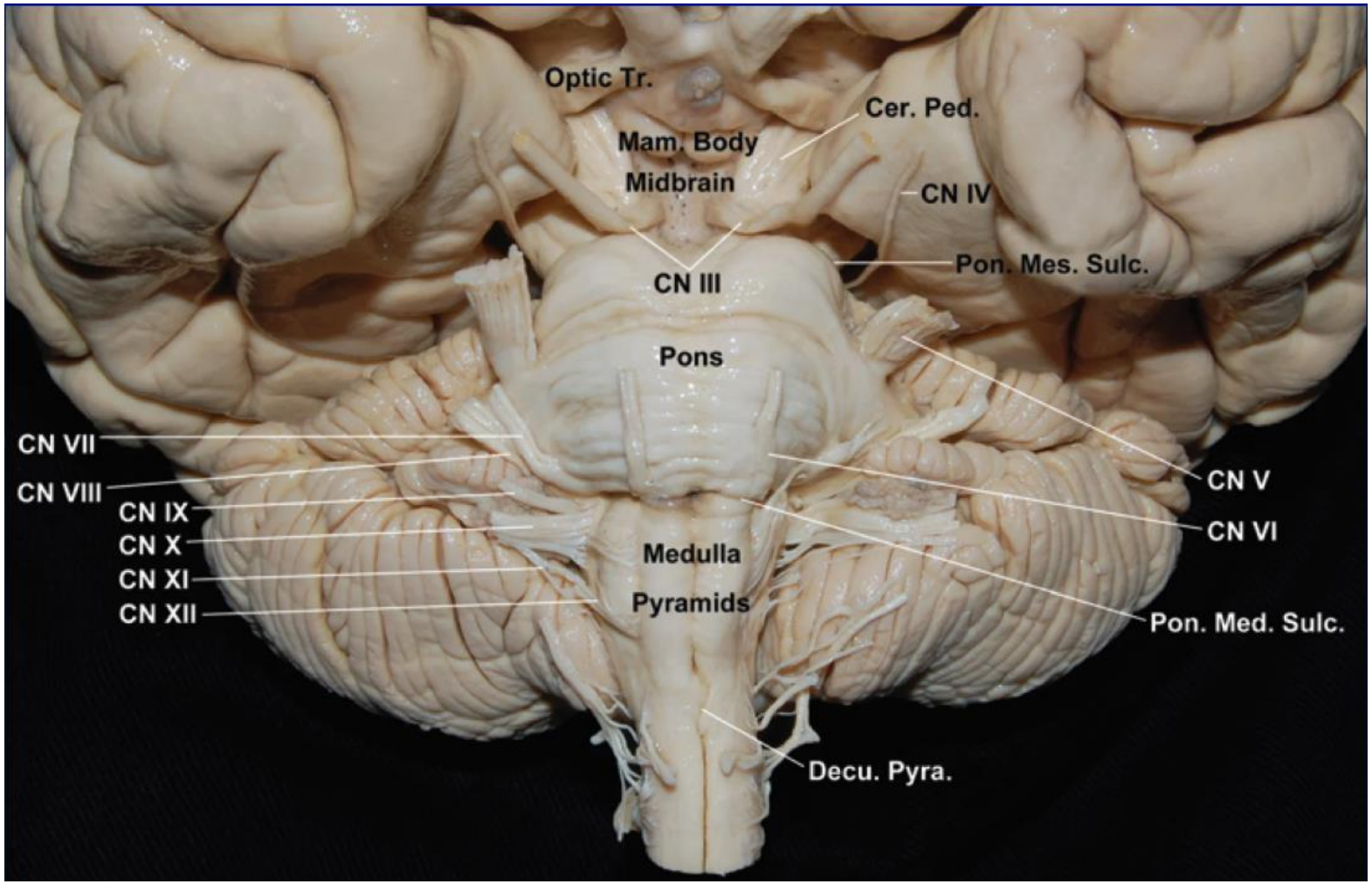
Midbrain:
- CN III
- CN IV exits dorsally

Pons: CN V

Pontomedullary Junction
- CN VI, VII, VIII

Medulla: CN IX, X, XII

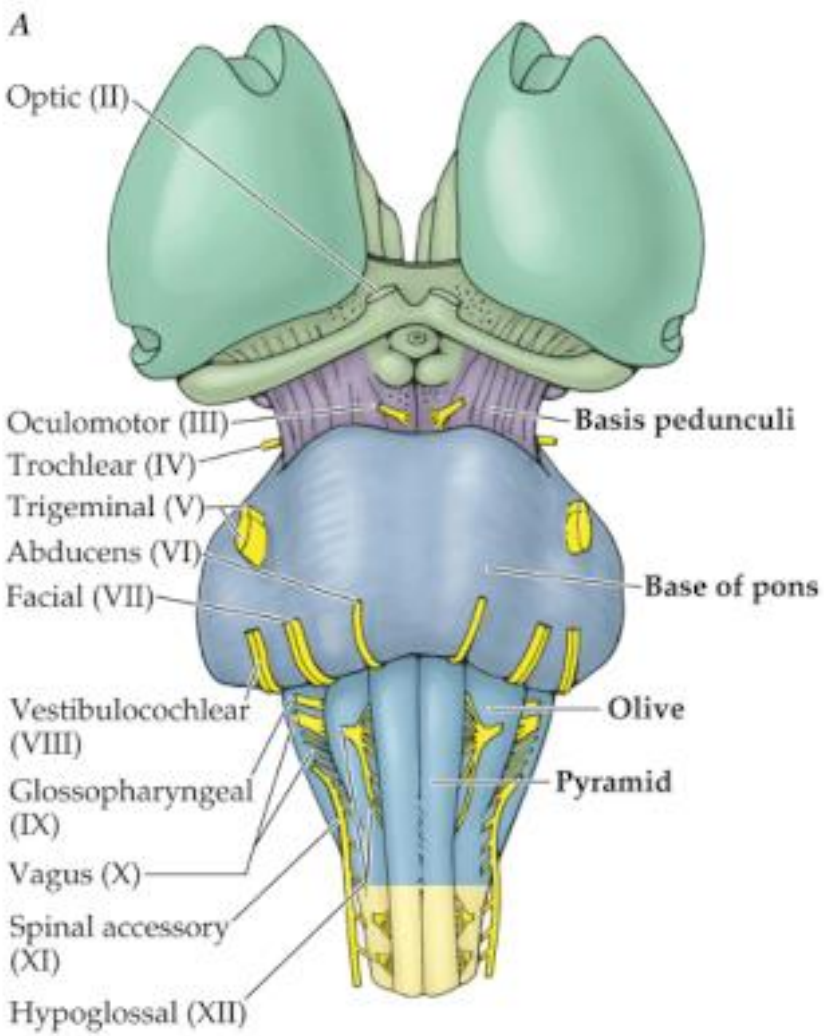
CN XI
- Spinal roots arise from upper cervical cord and ascend into skull



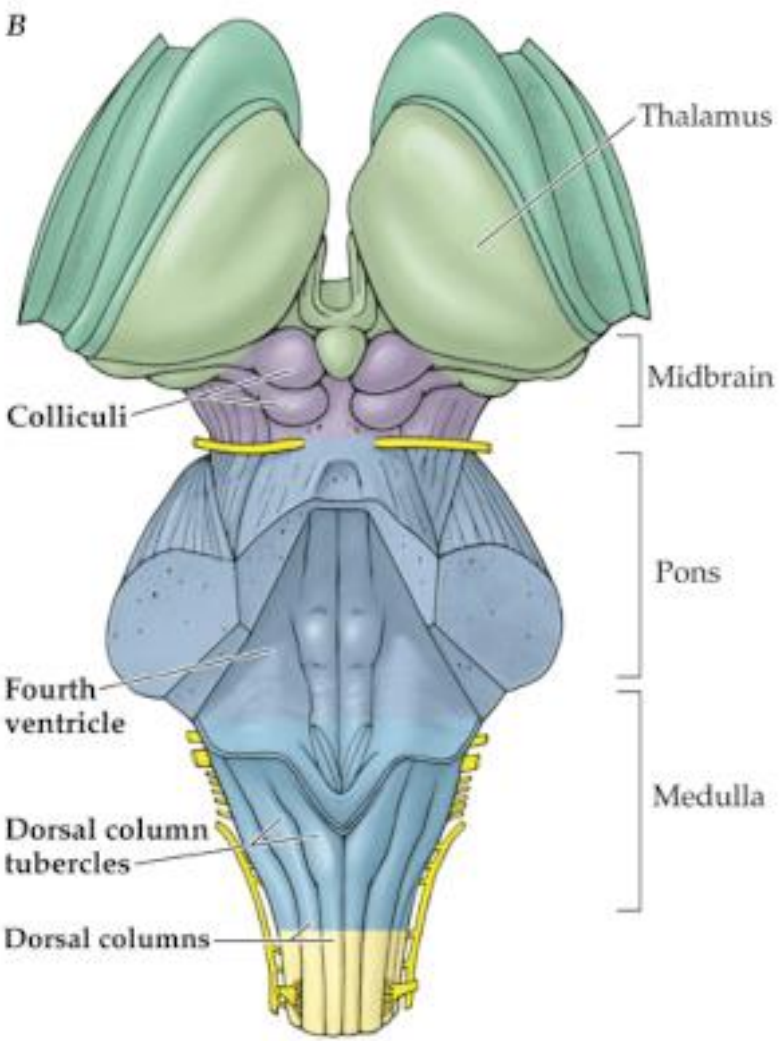
CN deficit helps identify the brainstem level

Cranial Nerves	
CN I*	Olfactory
CN II*	Optic
CN III	Oculomotor
CN IV	Trochlear
CN V	Trigeminal CN V1: Ophthalmic CN V2: Maxillary CN V3: Mandibular
CN VI	Abducent
CN VII	Facial
CN VIII	Vestibulocochlear
CN IX	Glossopharyngeal
CN X	Vagus
CN XI	Accessory
CN XII	Hypoglossal

*CN I and II are not brainstem cranial nerves.



Anterior View



Posterior View

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Image: NA Text & Atlas, Martin, 4e.

Blood Supply – Circle of Willis

Anterior Circulation

- Internal carotid artery → ACA, MCA

Posterior Circulation

- Vertebral arteries → PICA
- Vertebral arteries unite → Basilar artery
- Basilar artery branches → AICA, SCA, pontine aa.
- Basilar artery terminates as → PCA

Communicating Arteries

- Anterior communicating artery (ACOM)
- Posterior communicating arteries (PCOM)

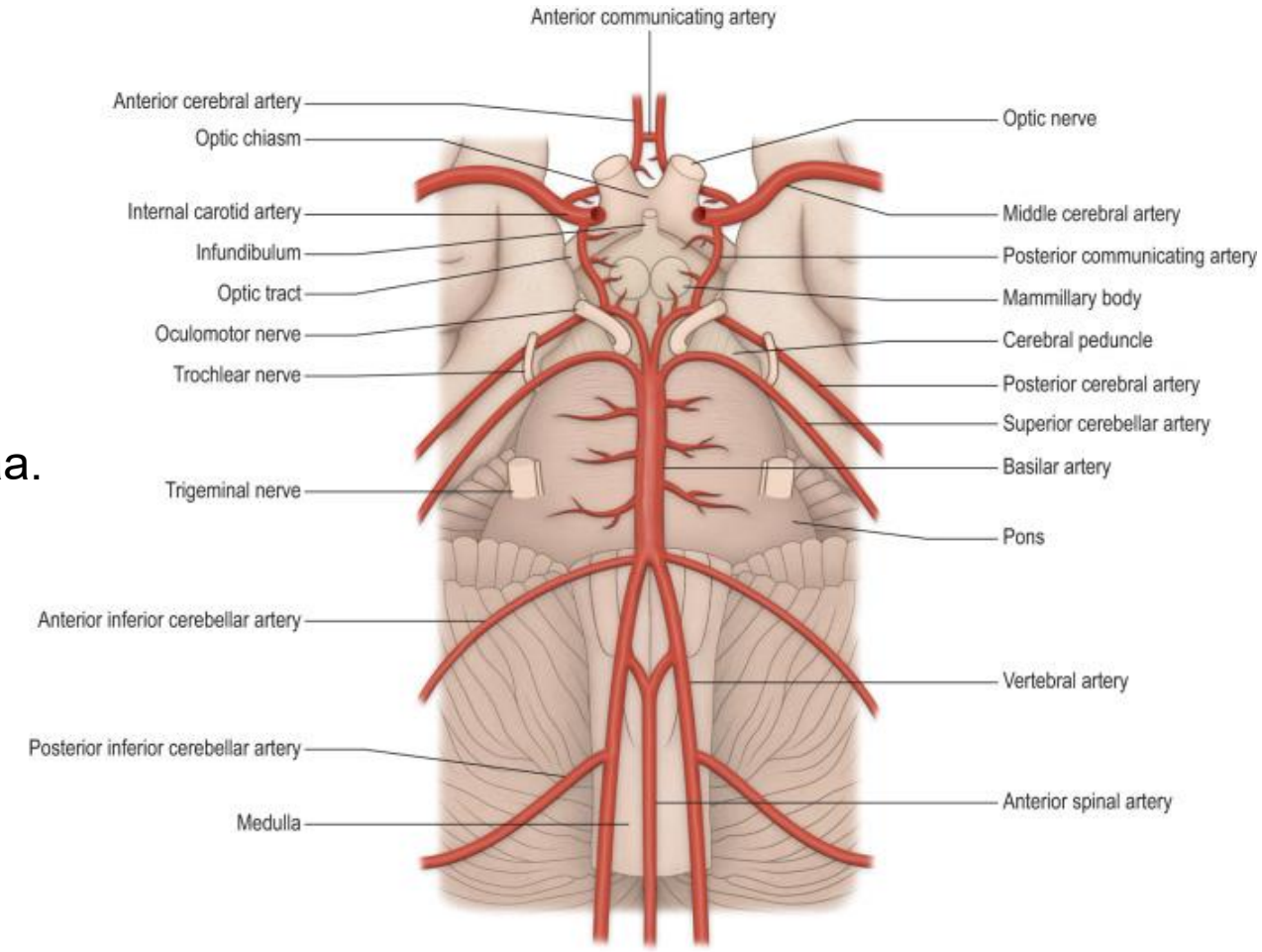


Fig. 7.2 The arrangement of arterial vessels on the base of the brain. The diagram shows the circulus arteriosus (circle of Willis).
Crossman, Neuroanatomy: An Illustrated Color Text, 6e.

Major Arterial Supply

Midbrain

- Posterior cerebral artery (PCA)
- Superior cerebellar artery (SCA)

Pons

- Pontine branches of basilar artery
- Anterior inferior cerebellar artery (AICA)
- Superior cerebellar artery (SCA)

Medulla

- Vertebral artery
- Posterior inferior cerebellar artery (PICA)
- Anterior spinal artery

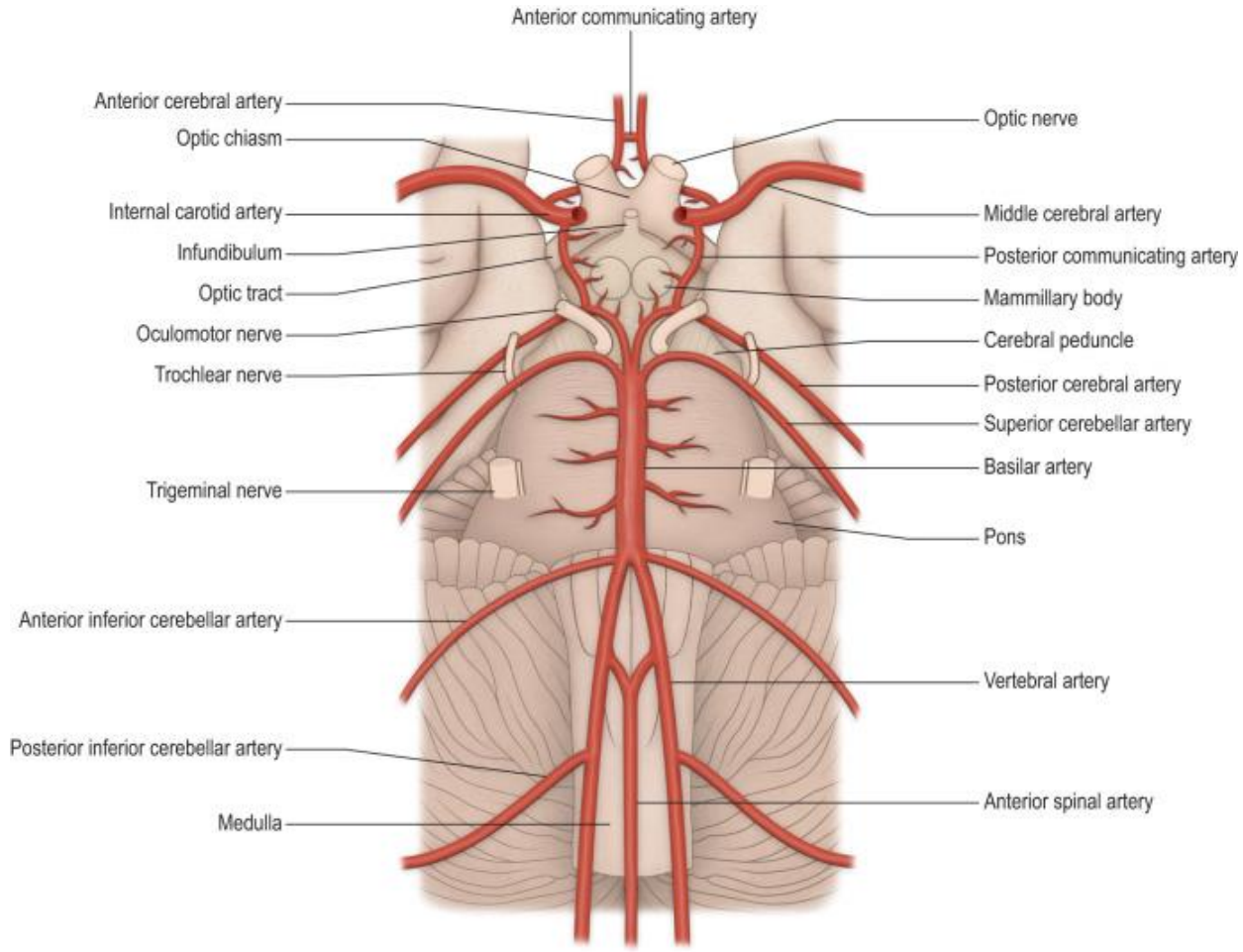


Fig. 7.2 The arrangement of arterial vessels on the base of the brain. The diagram shows the circulus arteriosus (circle of Willis).
Crossman, Neuroanatomy: An Illustrated Color Text, 6e.

Cranial Nerve Organization

Functional Components • Brainstem Nuclei • Pathways

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Cranial Nerve Functional Components

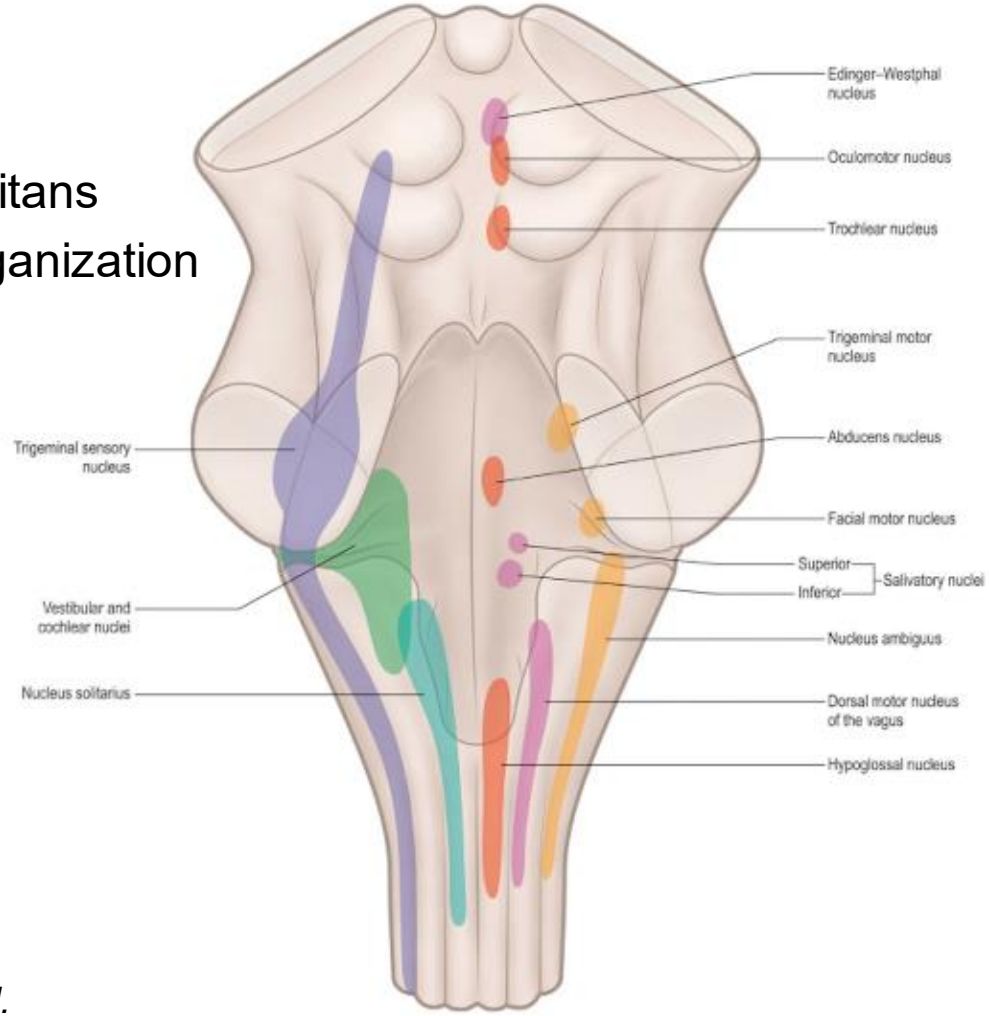
Abbreviation	Functional Component	General Function
	Motor	
GSE	General somatic efferent	Motor to skeletal muscles derived from somites
SVE	Special visceral efferent	Motor to pharyngeal arch muscles
GVE	General visceral efferent	Parasympathetic motor to glands, heart, and smooth muscle
	Sensory	
SVA	Special visceral afferent	Smell, taste
GVA	General visceral afferent	Sensory input from viscera
GSA	General somatic afferent	General sensations – touch, pain, temperature
SSA	Special somatic afferent	Vision, hearing, balance

Efferent = motor output | Afferent = sensory input

Organization of Cranial Nerve Nuclei

- Motor nuclei → medial
- Sensory nuclei → lateral
- Motor and sensory columns are separated by the sulcus limitans
- Functional columns follow a predictable medial-to-lateral organization

Motor = medial | Sensory = lateral



Dorsal brainstem view showing sensory (left) and motor (right) cranial nerve nuclei; matching colors indicate common embryologic origin.
Neuroanatomy. Crossman, 7e.

GVA + SVA share the solitary nucleus: SVA mainly rostral; GVA mainly caudal.

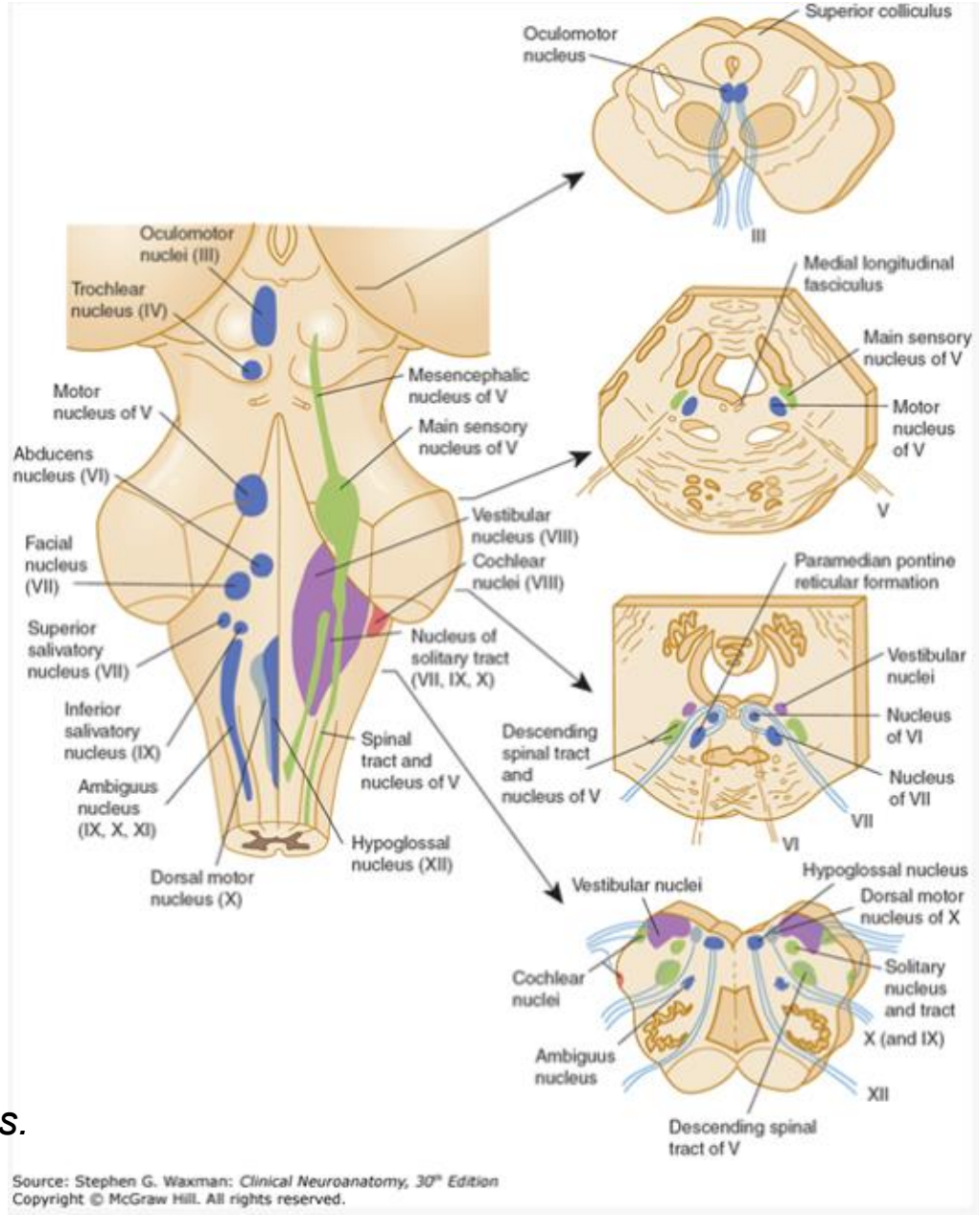
Cranial Nerve Nuclei by Level

Midbrain: III, IV

Pons: V, VI, VII, VIII

Medulla: IX, X, XII

Spinal Cord: XI



Source: Stephen G. Waxman: *Clinical Neuroanatomy, 30th Edition*
Copyright © McGraw Hill. All rights reserved.

Don't memorize every nucleus on this map. Use it to identify the brainstem level and recognize that some nuclei extend across levels.

Eye Movements & Ocular Motor Nerves

CN III • CN IV • CN VI • Extraocular Muscles • Peripheral Lesions

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Extraocular Muscle Innervation

- **CN III:** medial rectus, superior rectus, inferior rectus, inferior oblique
- **CN IV:** superior oblique
- **CN VI:** lateral rectus
- **CN III:** levator palpebrae superioris

LR6 • SO4 • All others III

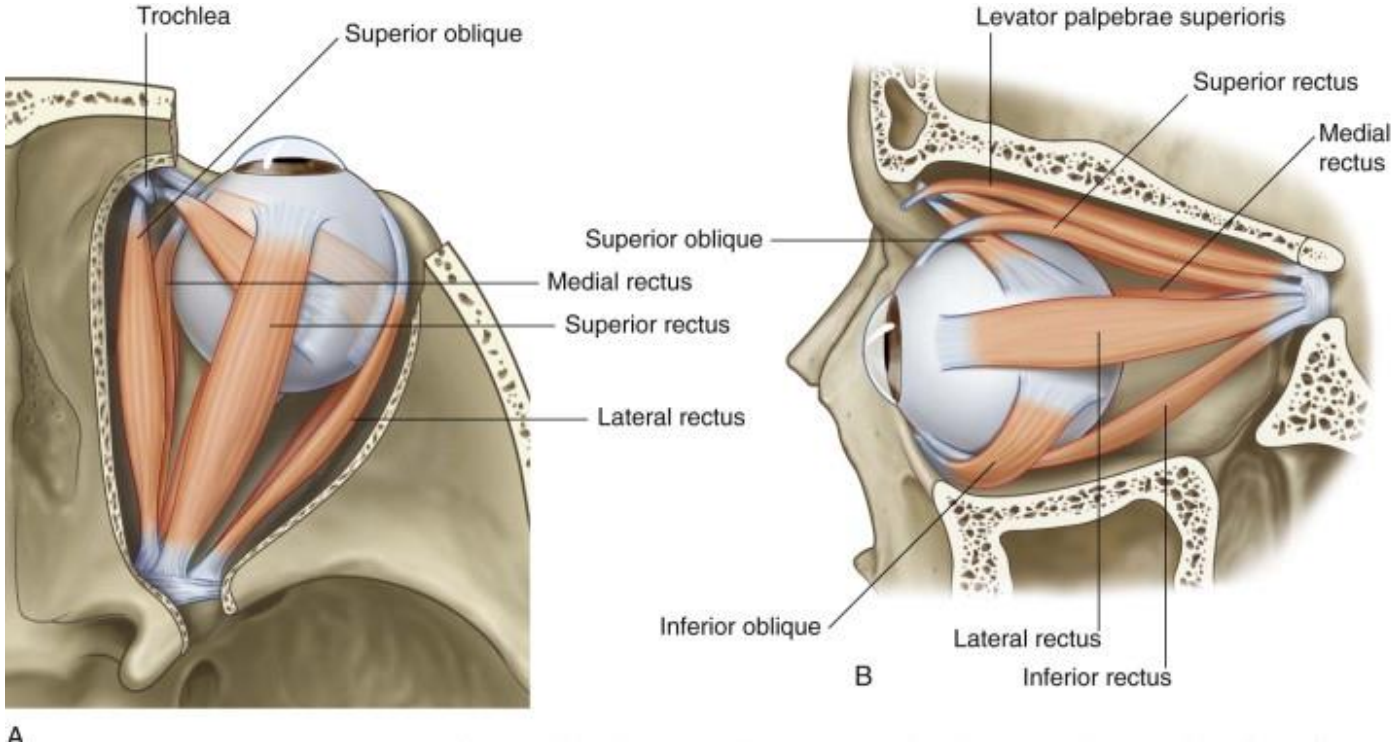
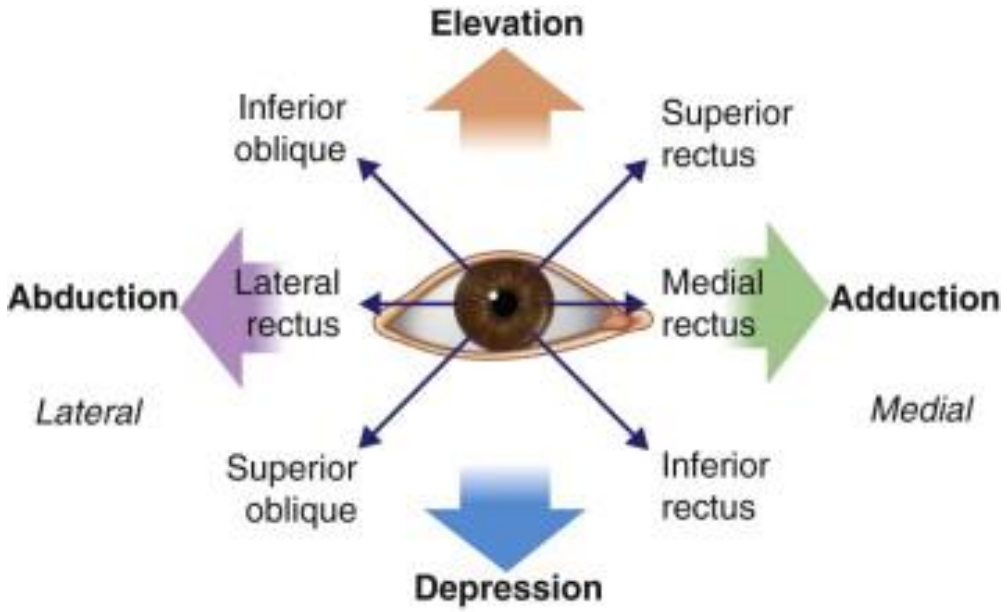


Fig. 8.92 Gray's Anatomy for Students, 3e.

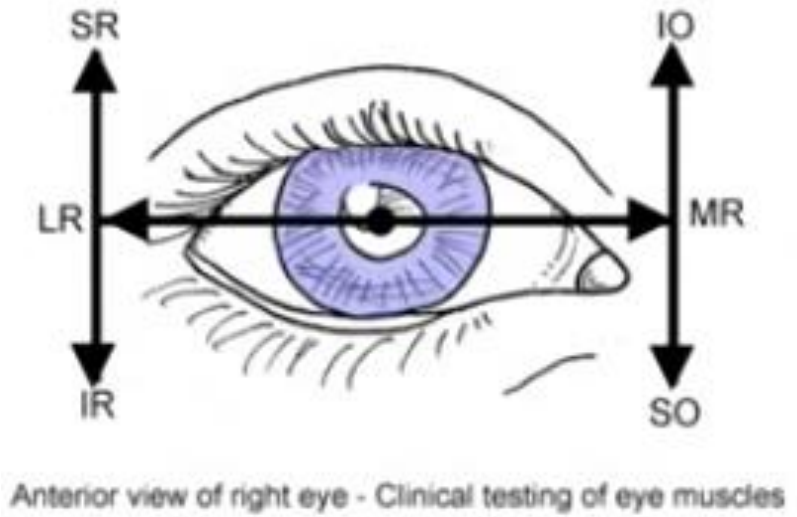
Anatomical Actions vs Clinical Testing

Watch the video:
https://youtu.be/3J2UZiLVZKA?feature=shared

- Extraocular muscles have multiple anatomical actions
- Clinical testing positions the eye to preferentially isolate a muscle's vertical action



Anatomical Actions



Anterior view of right eye - Clinical testing of eye muscles

Clinical Testing

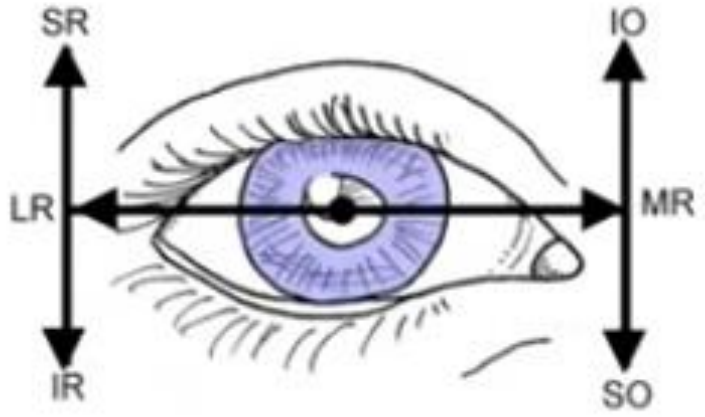
Recti: abduct first | Obliques: adduct first

Anatomical actions describe muscle biomechanics; clinical testing describes how muscle function is isolated on examination.

Extraocular Muscles: Actions & Testing

Muscle	Primary Clinical Test
Medial rectus	Adduct
Lateral rectus	Abduct
Superior rectus	**Abduct → elevate**
Inferior rectus	**Abduct → depress**
Superior oblique	**Adduct → depress**
Inferior oblique	**Adduct → elevate**

Recti tested in abduction | Obliques tested in adduction



Anterior view of right eye - Clinical testing of eye muscles

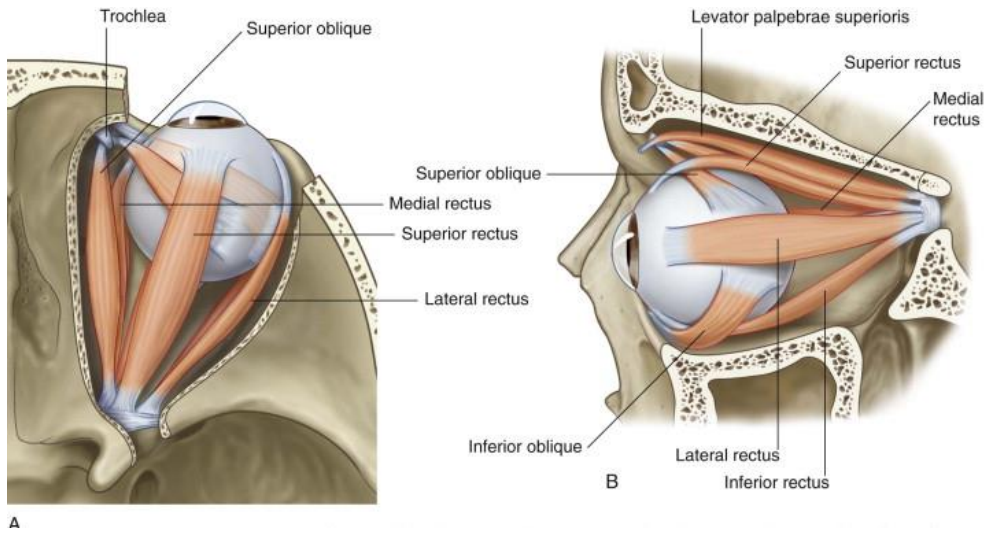


Fig. 8.92 Gray's Anatomy for Students, 3e.

Oculomotor Nerve (CN III)

- Origin: midbrain
- **Oculomotor nucleus (GSE)** → most extraocular muscles + levator palpebrae
- **Edinger–Westphal nucleus (GVE)** → parasympathetic fibers
- Passes through cavernous sinus → superior orbital fissure
- Parasympathetic: Edinger-Westphal → ciliary ganglion → short ciliary nerves
- Controls: sphincter pupillae → miosis | Ciliary muscle → accommodation

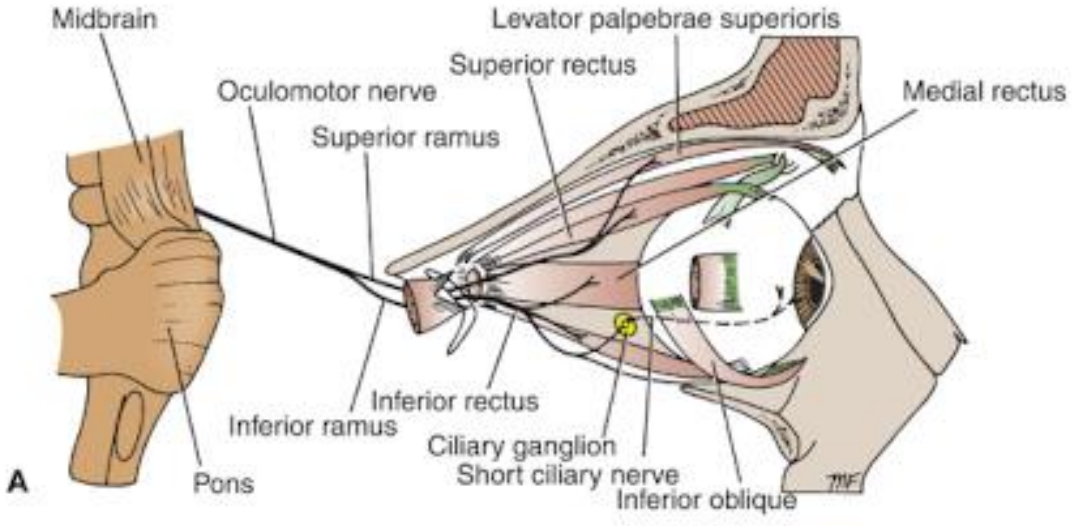


Fig. 12.61A. Origin and distribution of the oculomotor nerve. Wineski, SCABR, 10e.

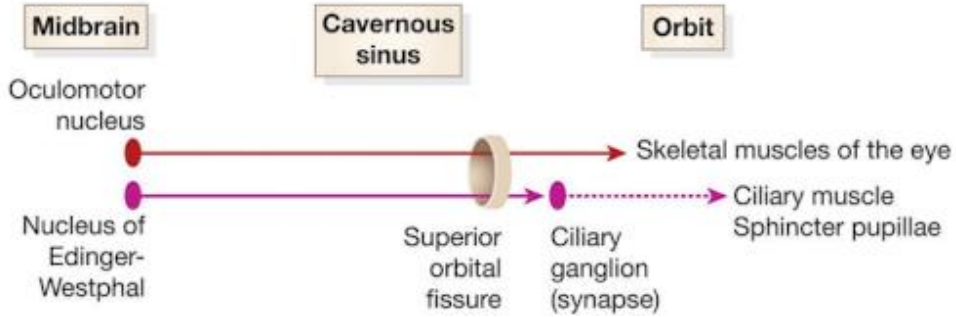


Fig. 12.61A. Origin and distribution of the oculomotor nerve. Wineski, SCABR, 10e. Champney. ECNA, 2016.

Oculomotor Nerve Palsy

- Eye “down and out”
- Ptosis
- Mydriasis
- Loss of accommodation
- Diplopia

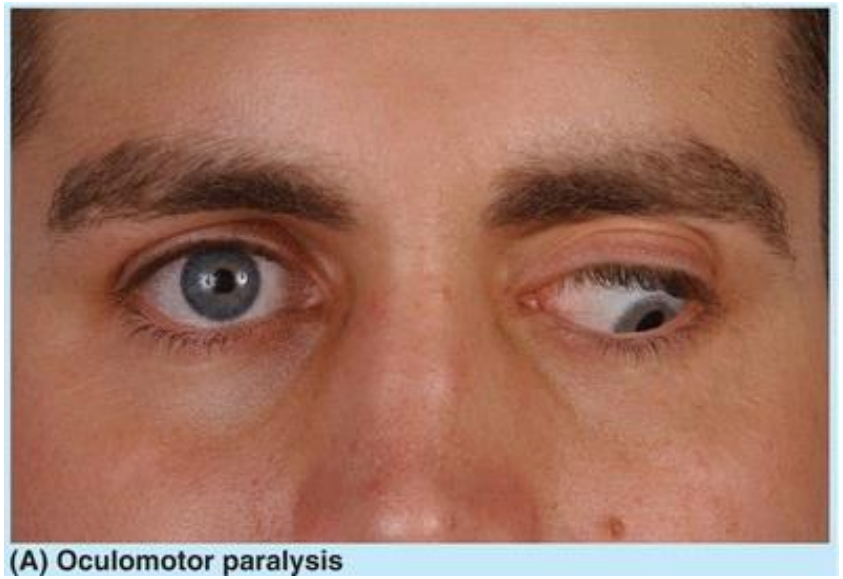


Fig. B8.31A Clinically Oriented Anatomy, *9e*.

Pupil Involvement

- Compressive lesion → pupil often involved (dilated)
- Ischemic lesion → pupil often spared

CN III palsy → Ptosis + 3Ds
(Down + Deviated + Dilated pupil)

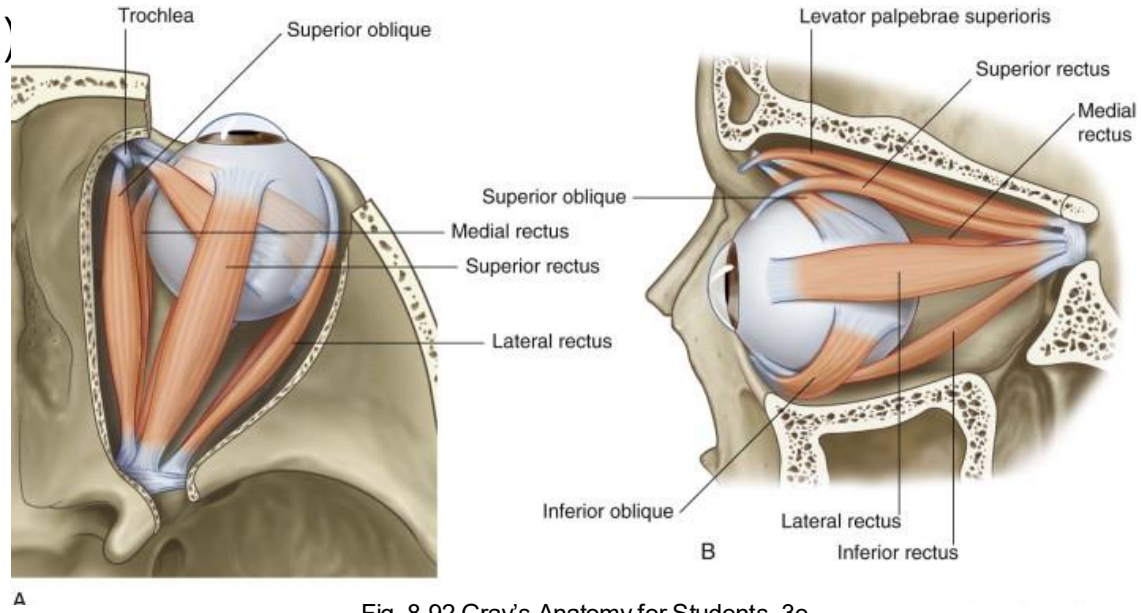


Fig. 8.92 Gray’s Anatomy for Students, 3e.

Trochlear Nerve (CN IV)

- Origin: midbrain
- **Motor (GSE)** → superior oblique
- Only CN that exits dorsally
- Fibers decussate before exiting
- Passes through cavernous sinus → superior orbital fissure
- Superior oblique: intorsion + depression in adduction

CN IV → superior oblique → "down and in" testing

CN IV nucleus lesion → contralateral superior oblique weakness
CN IV nerve lesion → ipsilateral superior oblique weakness

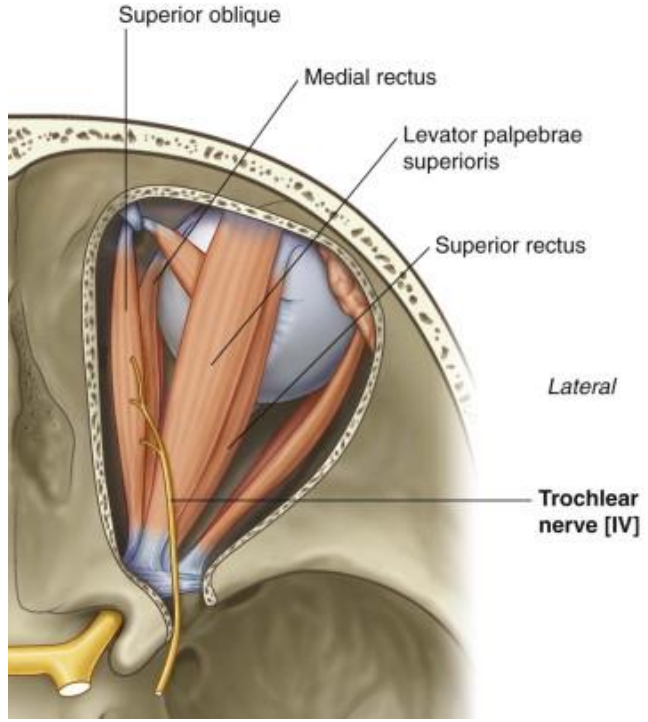
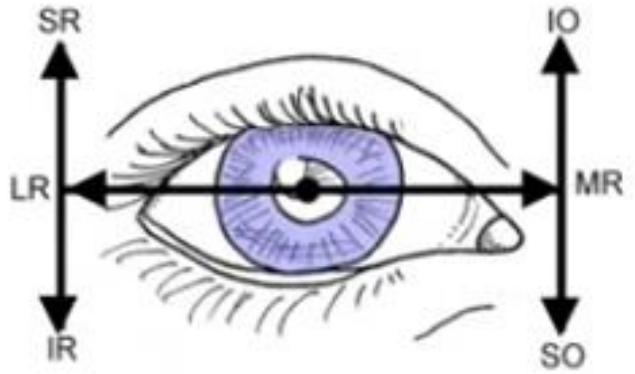


Fig. 8.99 Gray's Anatomy for Students, 3e.

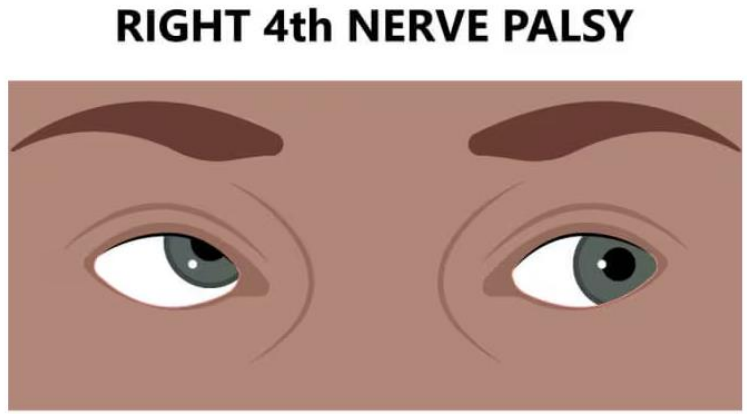


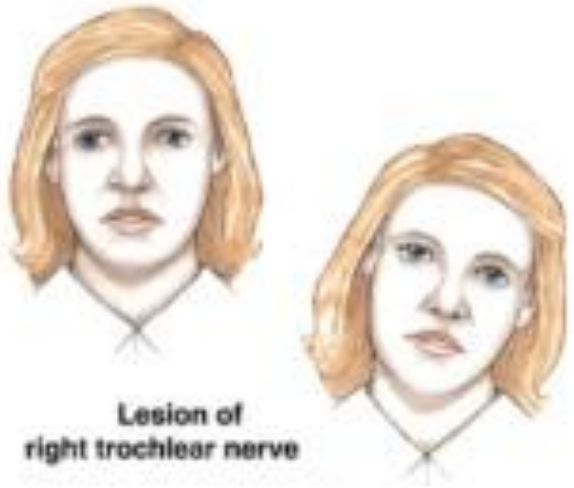
Anterior view of right eye - Clinical testing of eye muscles

Trochlear Nerve Palsy

- Vertical/torsional diplopia
- Worse looking down and in
- Difficulty descending stairs or reading
- Affected eye tends to be elevated and extorted
- Compensatory head tilt away from lesion

CN IV palsy → trouble looking down when the eye is adducted

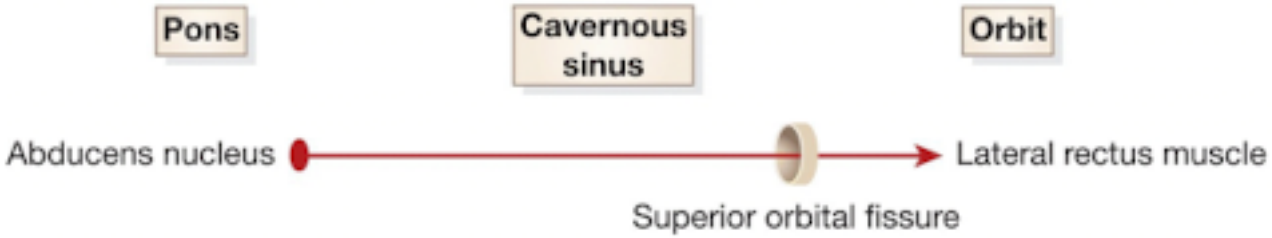




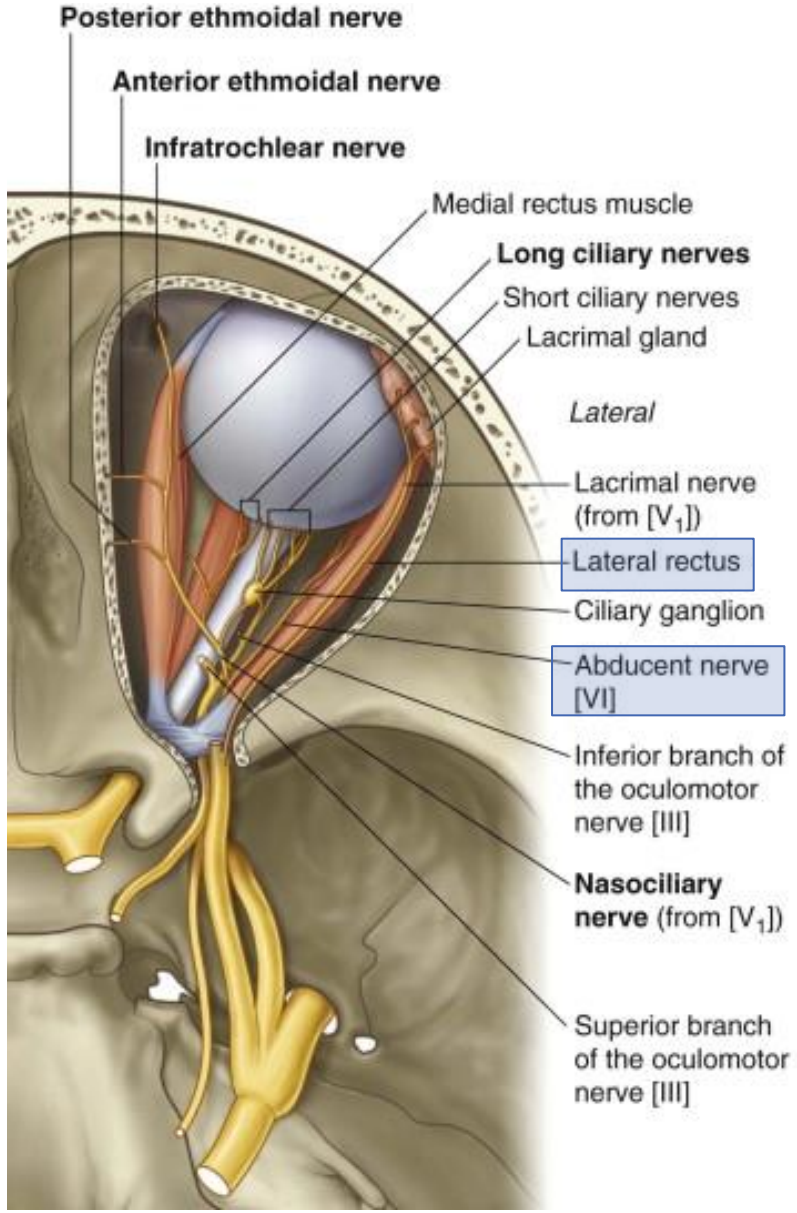
Affected eye drifts up and extorts. Individuals with trochlear nerve lesion compensate for this by tilting their head forward (chin tucked in) and to the side opposite the lesion.

Abducens Nerve (CN VI)

- Origin: pons
- Motor (GSE) → lateral rectus
- Passes through cavernous sinus → superior orbital fissure
- Lateral rectus abducts the eye



CN VI → lateral rectus → abduction



Abducens Nerve Palsy

- Lateral rectus paralysis
- Eye pulled medially (esotropia)
- Horizontal diplopia
- Worse looking toward the affected side
- Compensatory head turn toward lesion



Fig. B8.31B Clinically Oriented Anatomy, *9e*.

CN VI palsy → cannot abduct the affected eye

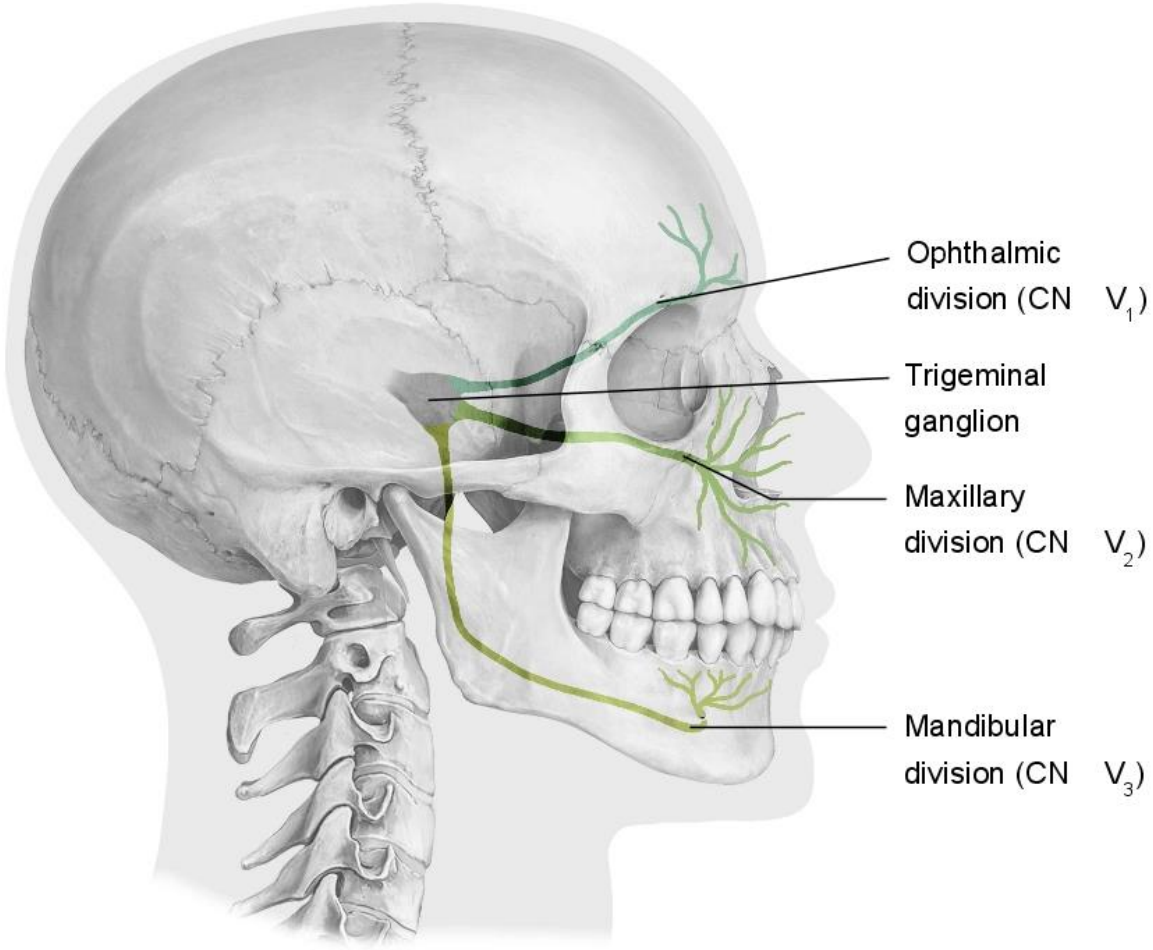
Facial Sensation & Movement

CN V • CN VII • Facial Sensation • Facial Movement • Corneal Reflex

Trigeminal Nerve (CN V)

- Origin: pons
- Major sensory nerve of the face
- **V1 Ophthalmic**: sensory
- **V2 Maxillary:** sensory
- **V3 Mandibular:** sensory + branchial motor (SVE
- V3 motor → muscles of mastication

V1 + V2 = sensory | V3 = sensory + motor



©2008 Thieme Publishers

Illustrator: Karl Wesker

Trigeminal Divisions & Skull Foramina

- **V1 Ophthalmic** → superior orbital fissure
- **V2 Maxillary** → foramen rotundum
- **V3 Mandibular** → foramen ovale

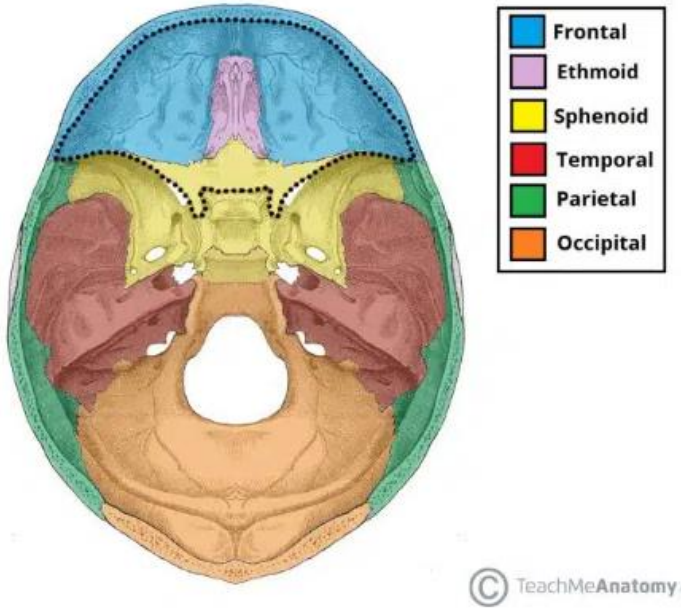


Fig 1

The bones of the base of the skull. The anterior cranial fossa has been outlined.

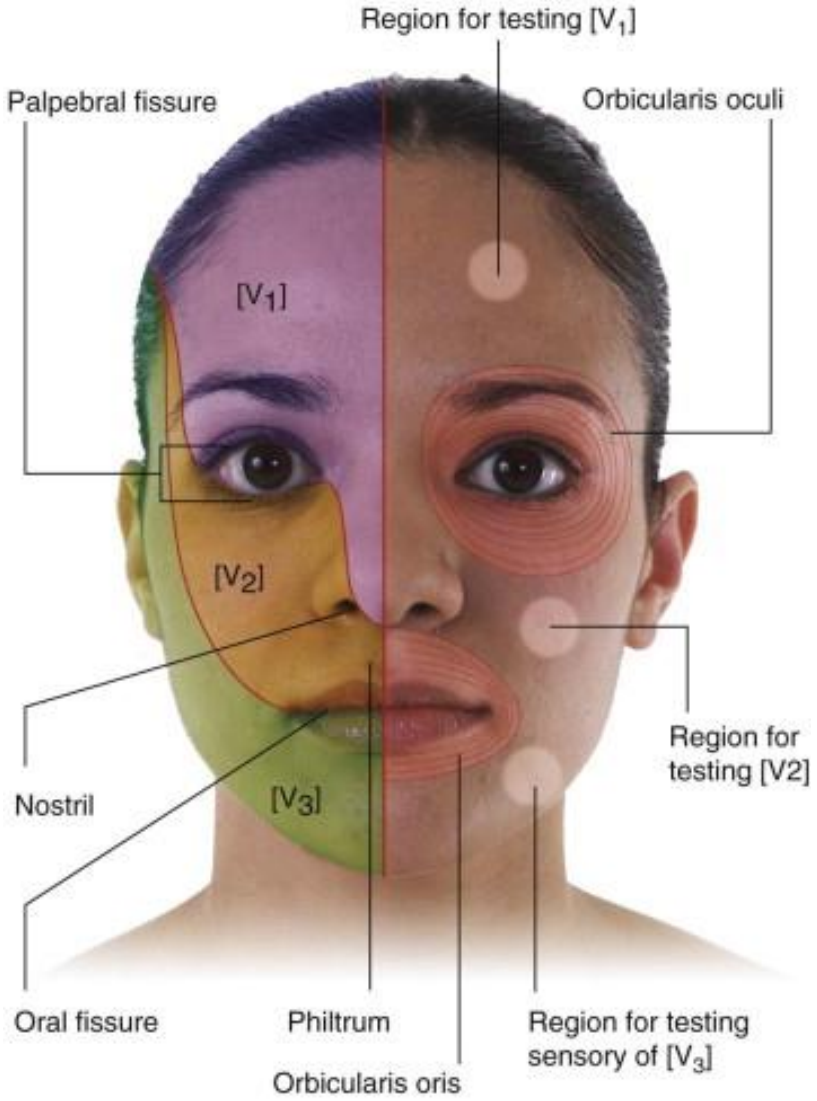
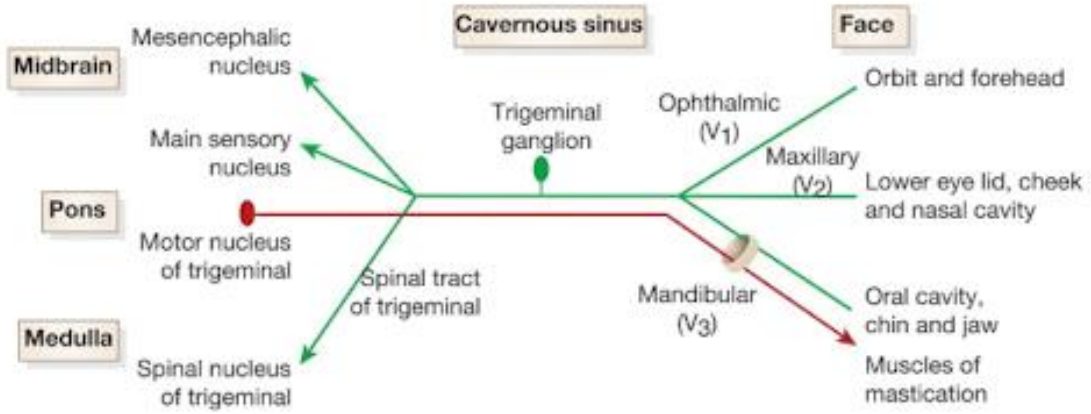
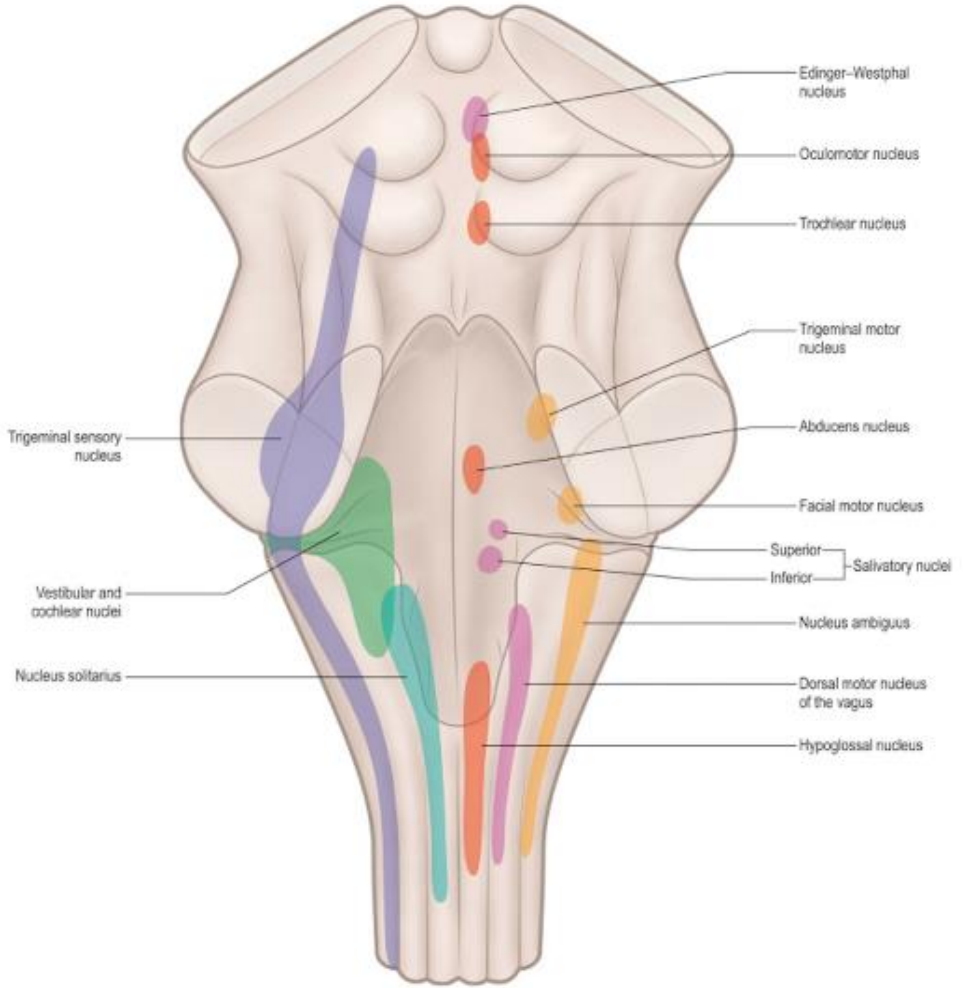


Fig 8.284. Gray's Anatomy for Students, 3e.

Trigeminal Sensory Nuclei

- **Mesencephalic nucleus:** proprioception
- **Principal (chief) sensory nucleus:** fine touch & vibration
- **Spinal trigeminal nucleus:** pain & temperature





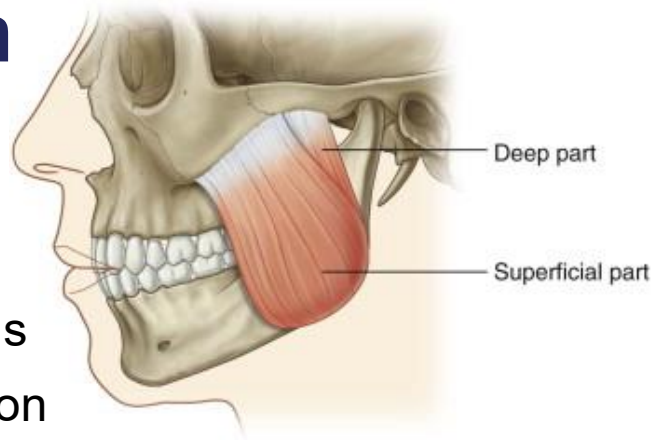
Midbrain → proprioception | Pons → touch | Caudal → pain & temperature

Champney. ECNA, 2016.
Dorsal brainstem view of the cranial nerve nuclei. Crossman, 7e.

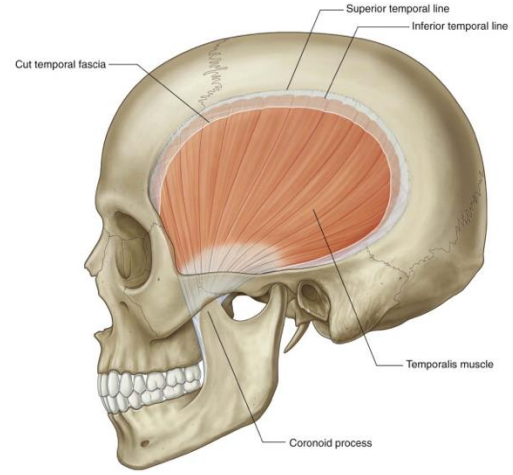
Trigeminal Motor Function

- **Motor nucleus of CN V:** pons
- V3 (SVE) → muscles of mastication
- Masseter, temporalis, medial & lateral pterygoids
- LMN lesion → weakness of ipsilateral mastication
- Jaw deviates toward the weak side on opening

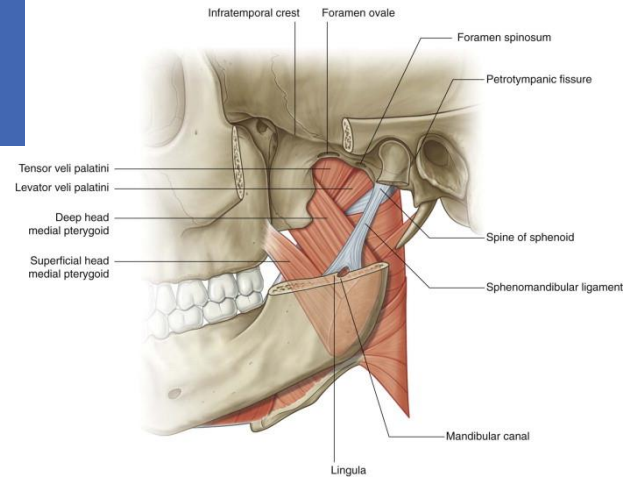
V3 LMN Lesion →
jaw deviates toward the weak side on opening



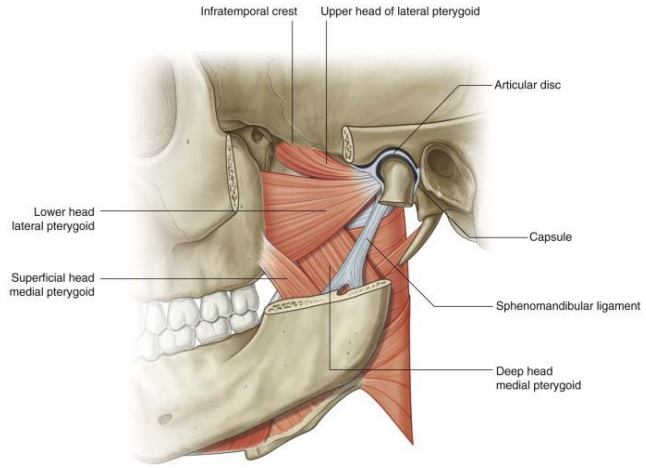
Masseter



Temporalis



Medial Pterygoid



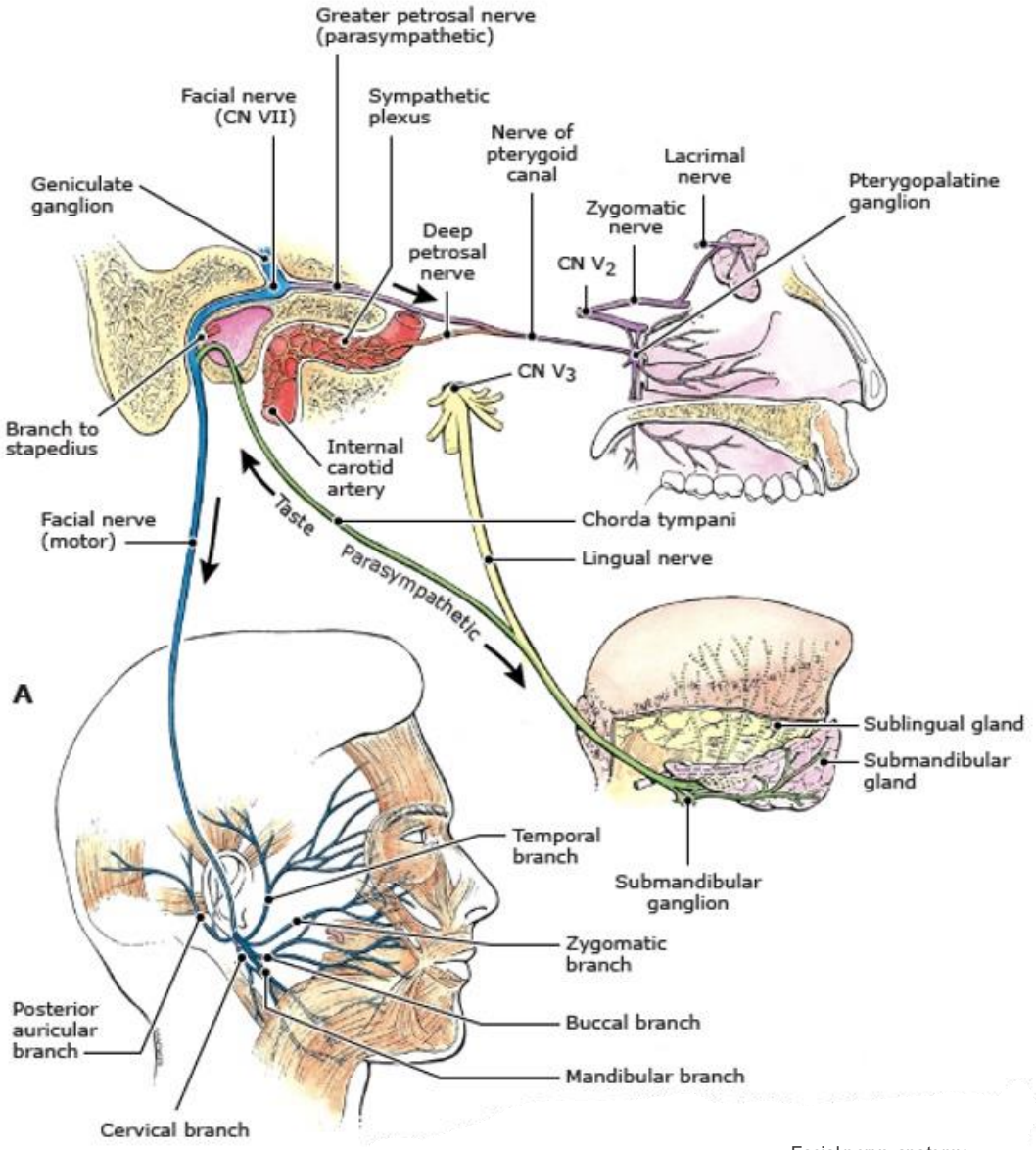
Lateral Pterygoid

Fig. 8.137, 8.139, 8.142, 8.143. GAS, 3e.

Facial Nerve (CN VII)

- Origin: pons
- **SVE:** muscles of facial expression
- **GVE:** lacrimal, submandibular & sublingual glands
- **SVA:** taste from anterior 2/3 of tongue
- **GSA:** small area of external ear
- Travels through internal acoustic meatus → facial canal
- Exits skull through stylomastoid foramen

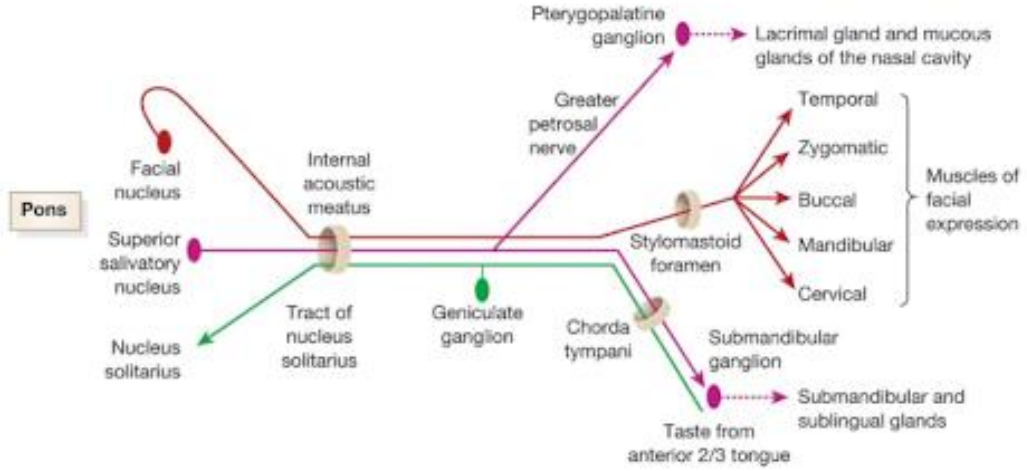
CN VII = facial movement + parasympathetic + taste

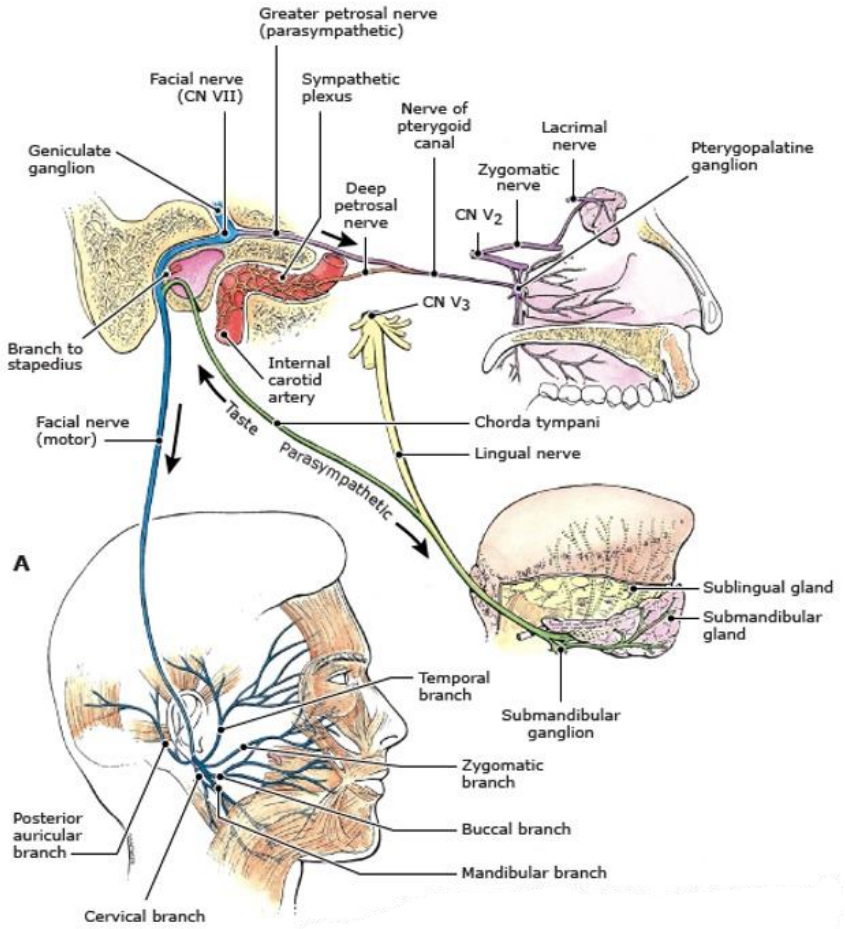


Facial nerve anatomy. UpToDate, 2019.

Facial Nerve: Major Branches

- **Greater petrosal nerve** → lacrimal/nasal/palatal glands
- **Nerve to stapedius**
- **Chorda tympani** → taste anterior 2/3 + salivary parasympathetics
- Motor branches → **muscles of facial expression**





Lesion location determines which CN VII functions are lost

Facial nerve anatomy. UpToDate, 2019.

Motor Innervation

Facial Nerve

- Temporal branch
- Zygomatic branch
- Buccal branch
- Marginal mandibular branch
- Cervical branch

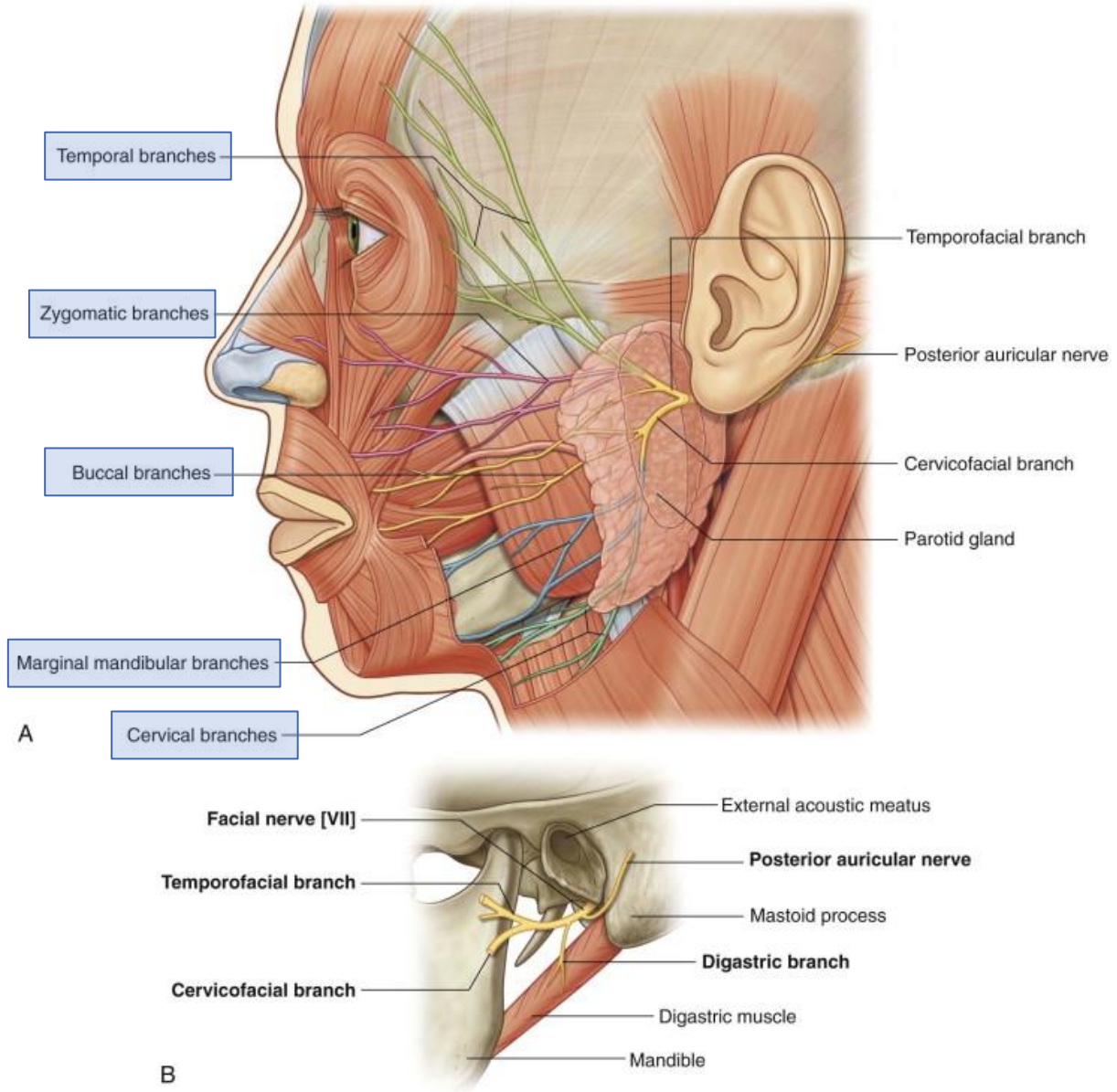


Fig. 8.63 Facial nerve on the face.
Gray's Anatomy for Students, 3e.

Muscles of the Face

Orbital Group
Orbital oculi
Corrugator supercilia

Nasal Group
Procerus
Nasalis
Depressor septi nasi

Oral Group
Buccinator
Orbicularis oris
Levator labii superioris alaeque nasi
Levator labii superioris
Zygomaticus minor
Zygomaticus major
Levator anguli oris
Risorius
Depressor anguli oris
Depressor labii inferioris
Mentalis

Other Muscles
Platysma
Auricular muscles
Occipitofrontalis

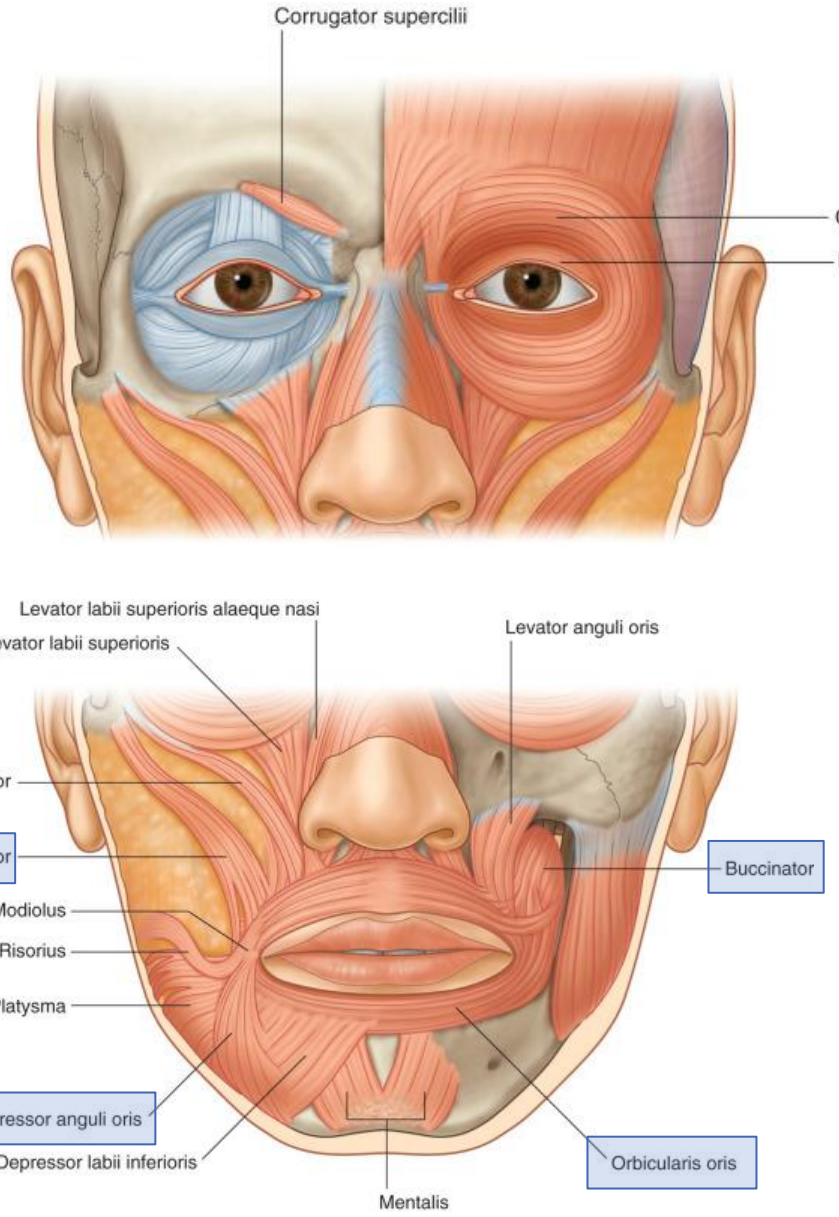


Fig. 8.54, 8.56 Gray's Anatomy for Students, 3e.

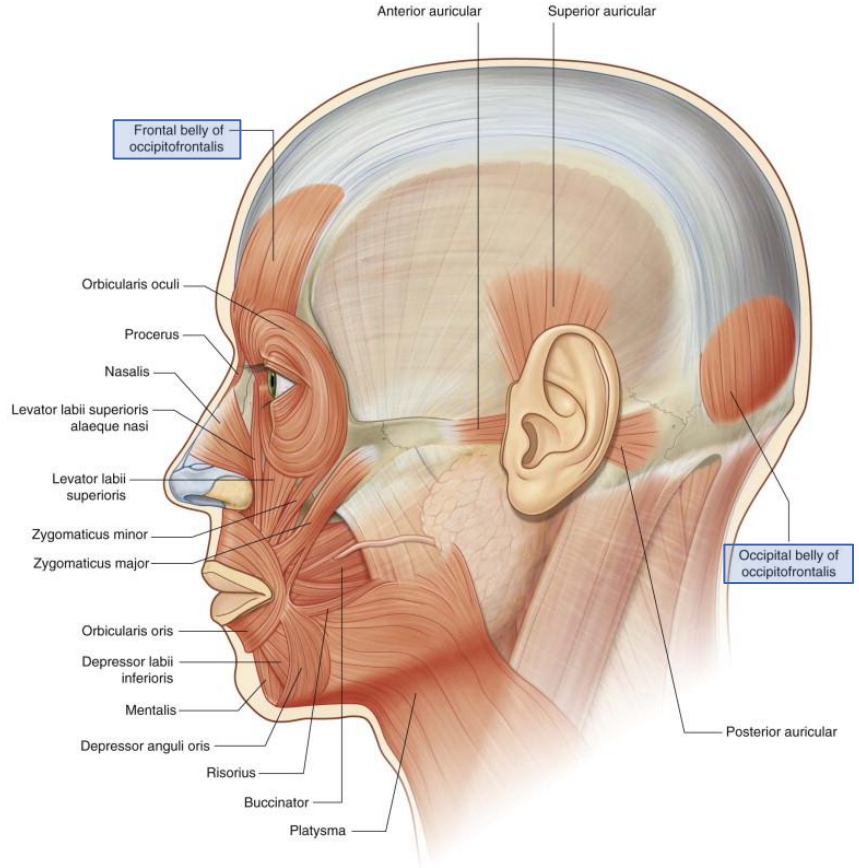


Fig. 8.53 Facial muscles.
Gray's Anatomy for Students, 3e.

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Central vs Peripheral Facial Weakness

Corticobulbar Innervation

- Most cranial motor nuclei receive bilateral cortical input
- Important exceptions: lower facial motor neurons (CN VII) and genioglossus motor neuron (CN XII) receive predominantly contralateral input

UMN Lesion

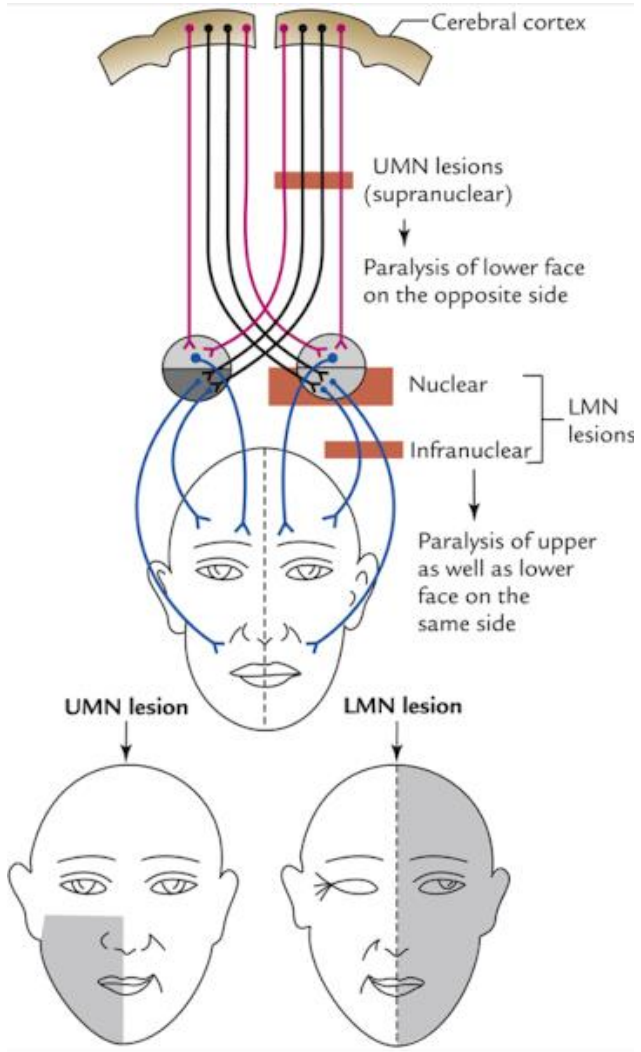
- Contralateral lower-face weakness
- Forehead relatively spared
- Upper face receives bilateral cortical input

LMN / CN VII Lesion

- Ipsilateral upper + lower-face weakness
- Forehead not spared



Bell Palsy



Forehead spared → UMN | Entire ipsilateral face weak → LMN

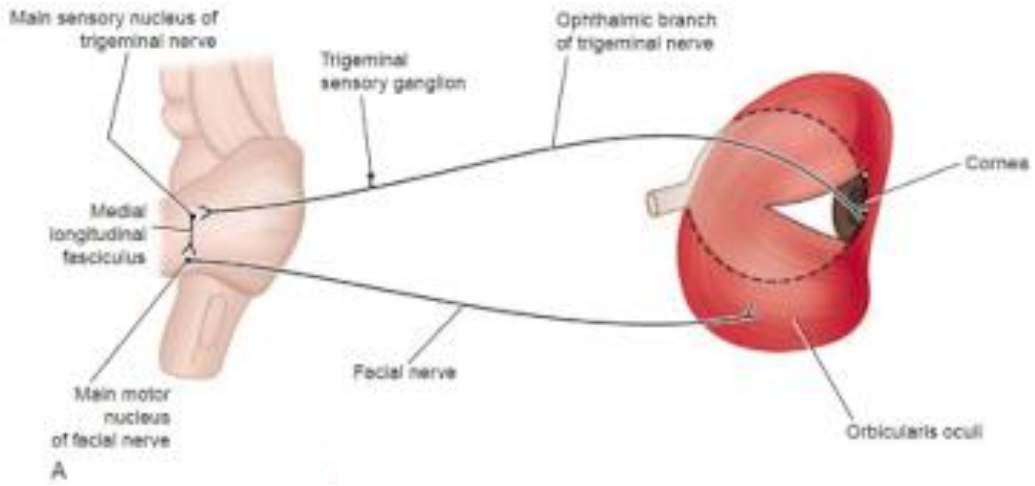
Bell palsy = peripheral CN VII (LMN) palsy

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Singh. TCNA, 3e.

Corneal (Blink) Reflex

- Corneal touch activates V1 (nasociliary branch)
- **Afferent limb:** CN V1
- **Efferent limb:** CN VII
- CN VII → orbicularis oculi → eyelid closure
- Normal response: bilateral blink



Corneal reflex: V1 in → VII out

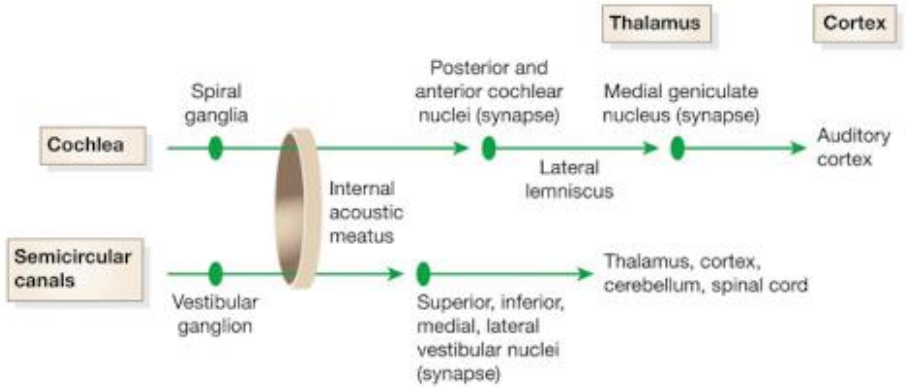
Figure 11.4 A: Corneal reflex. Splittgerber, SCNA, 8e.

Hearing & Balance

CN VIII • Cochlear Division • Vestibular Division • Clinical Correlates

Vestibulocochlear Nerve (CN VIII)

- Origin/attachment: pontomedullary junction
- Special somatic afferent (SSA)
- Cochlear division → hearing
- Vestibular division → balance & head movement
- Enters internal acoustic meatus (IAM) with CN VII
- Sensory nuclei extend across the pons–medulla region



CN VIII = SSA → hearing + balance

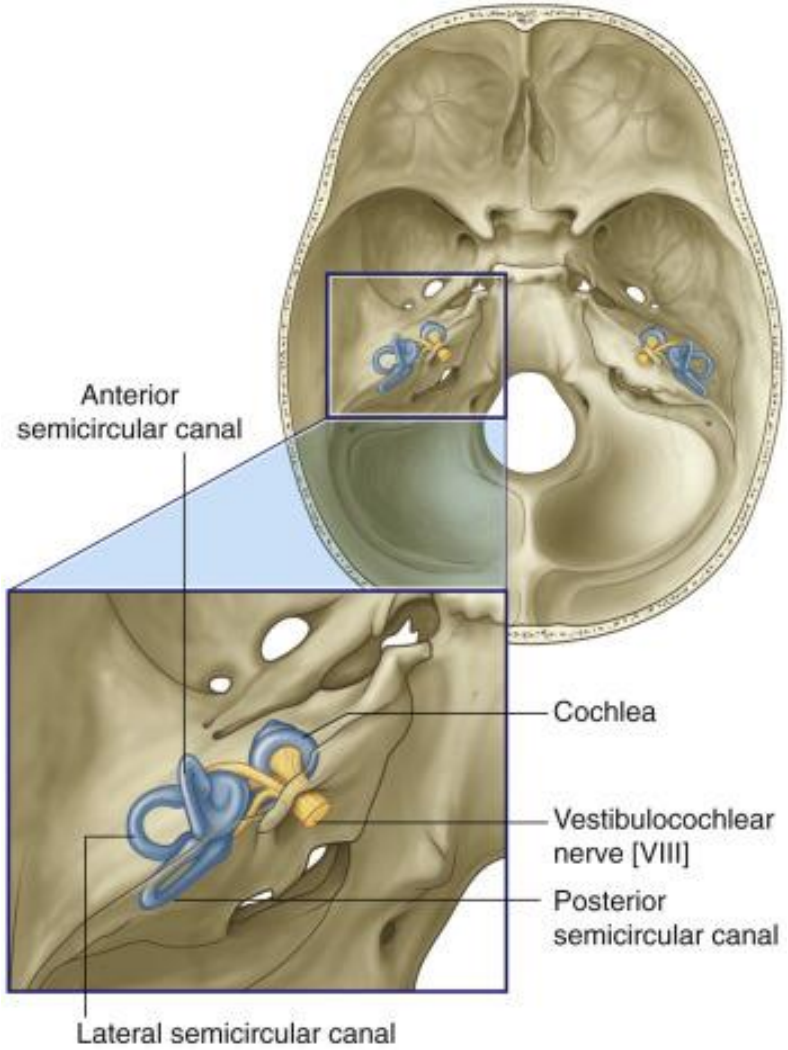


Fig. 8.123 Gray's Anatomy for Students, 3e.

CN VIII: Cochlear vs Vestibular

Cochlear Division

- Sensory input from the cochlea
- Function → hearing
- Terminates in cochlear nuclei
- Lesion → sensorineural hearing loss ± tinnitus

Vestibular Division

- Sensory input from the vestibular apparatus
- Function → balance & head movement
- Terminates in vestibular nuclei
- Lesion → vertigo, nystagmus, imbalance

Cochlear → hearing | Vestibular → balance

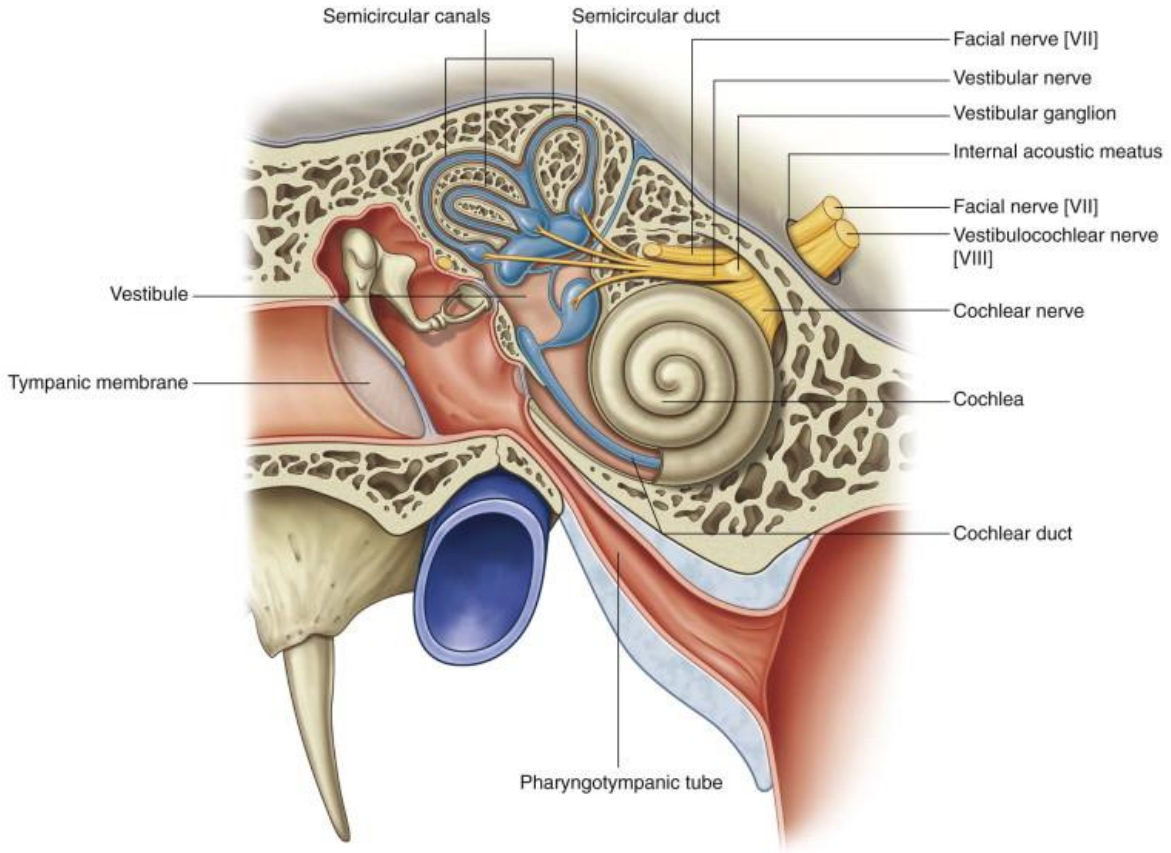
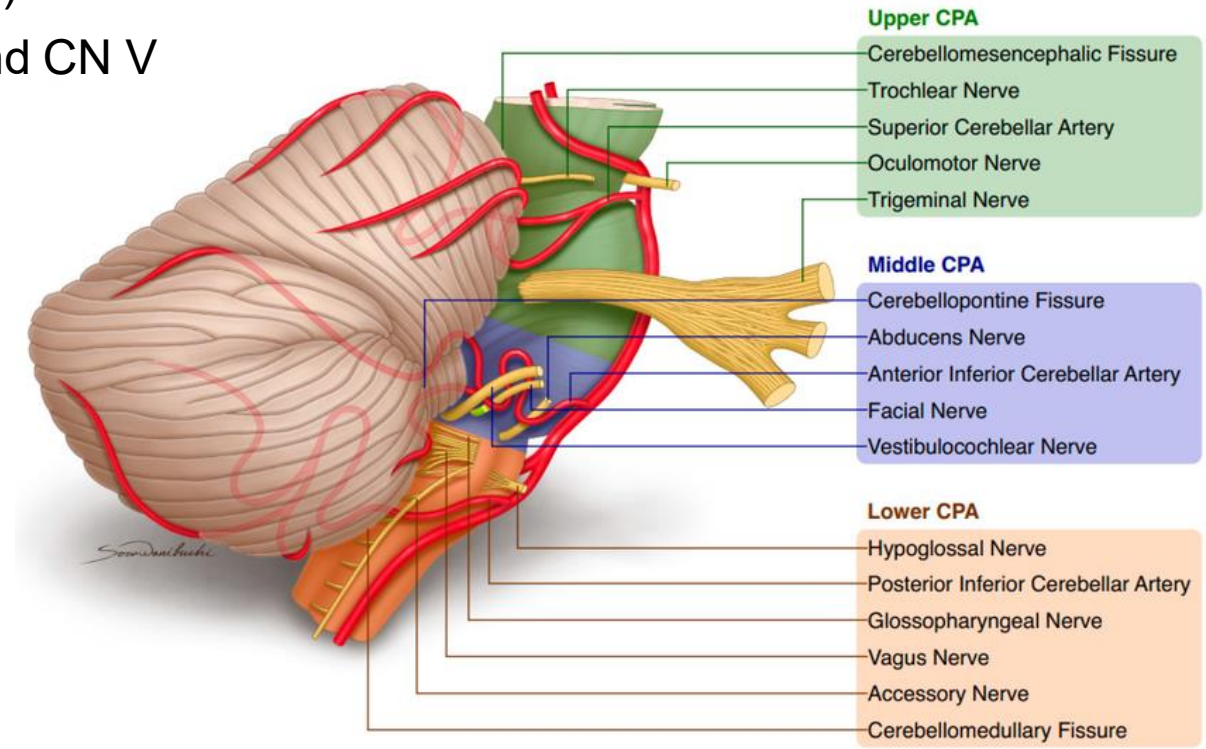


Fig. 8.124 Gray's Anatomy for Students, 3e.

Vestibular Schwannoma

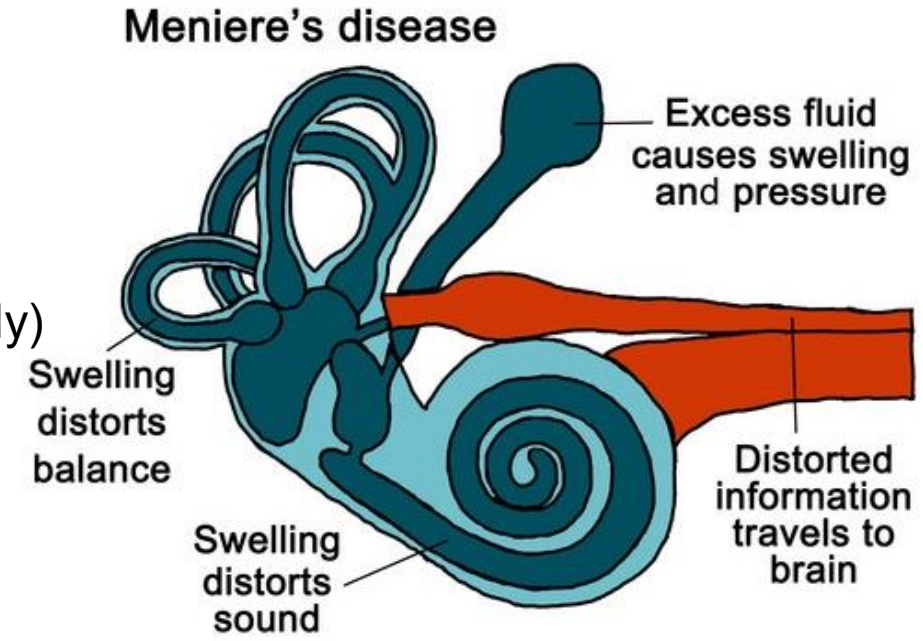
- Schwann-cell tumor, usually arising from the vestibular division of CN VIII
- Often begins within the internal acoustic meatus
- CN VIII: unilateral sensorineural hearing loss, tinnitus, imbalance
- May extend into the cerebellopontine angle (CPA)
- Larger tumors may compress adjacent CN VII and CN V
- Acoustic neuroma is a historical misnomer



The anatomy of cerebellopontine angle (CPA).
Kościołek, *Biomedicines*. 2024

Ménière's Disease (Endolymphatic Hydrops)

- Disorder of the inner ear caused by excess endolymph in the membranous labyrinth.
- Increased endolymph pressure → distention of cochlear and vestibular membranes → dysfunction of hair cells.



Clinical Triad

- Vertigo (recurrent, episodic)
- Tinnitus (ringing or roaring in the ear)
- Fluctuating sensorineural hearing loss (usually unilateral early)

Additional Features

- Sensation of ear fullness or pressure
- Symptoms may be triggered by stress, tiredness (fatigue), illness, dietary factor

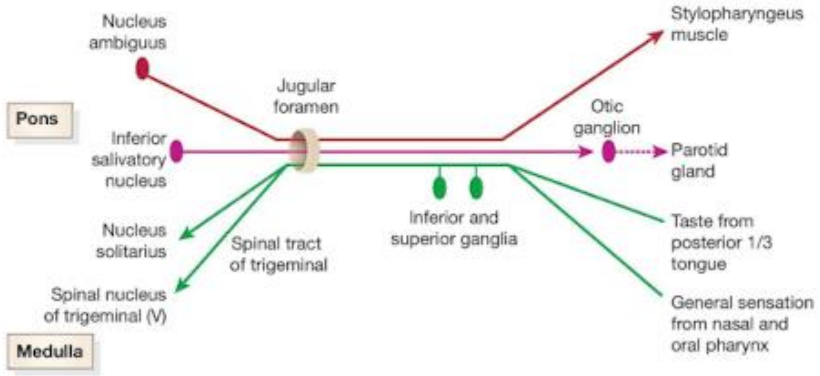
Image: Medline Plus

Pharynx, Larynx & Visceral Function

CN IX • CN X • Swallowing • Phonation • Gag Reflex • Visceral Function

Glossopharyngeal Nerve (CN IX)

- Origin: medulla
- **SVE:** stylopharyngeus → swallowing
- **GVE:** parotid gland
- **SVA:** taste from posterior 1/3 of tongue
- **GVA:** carotid sinus & carotid body
- **GSA:** posterior 1/3 tongue, oropharynx & middle ear
- Exits skull through jugular foramen



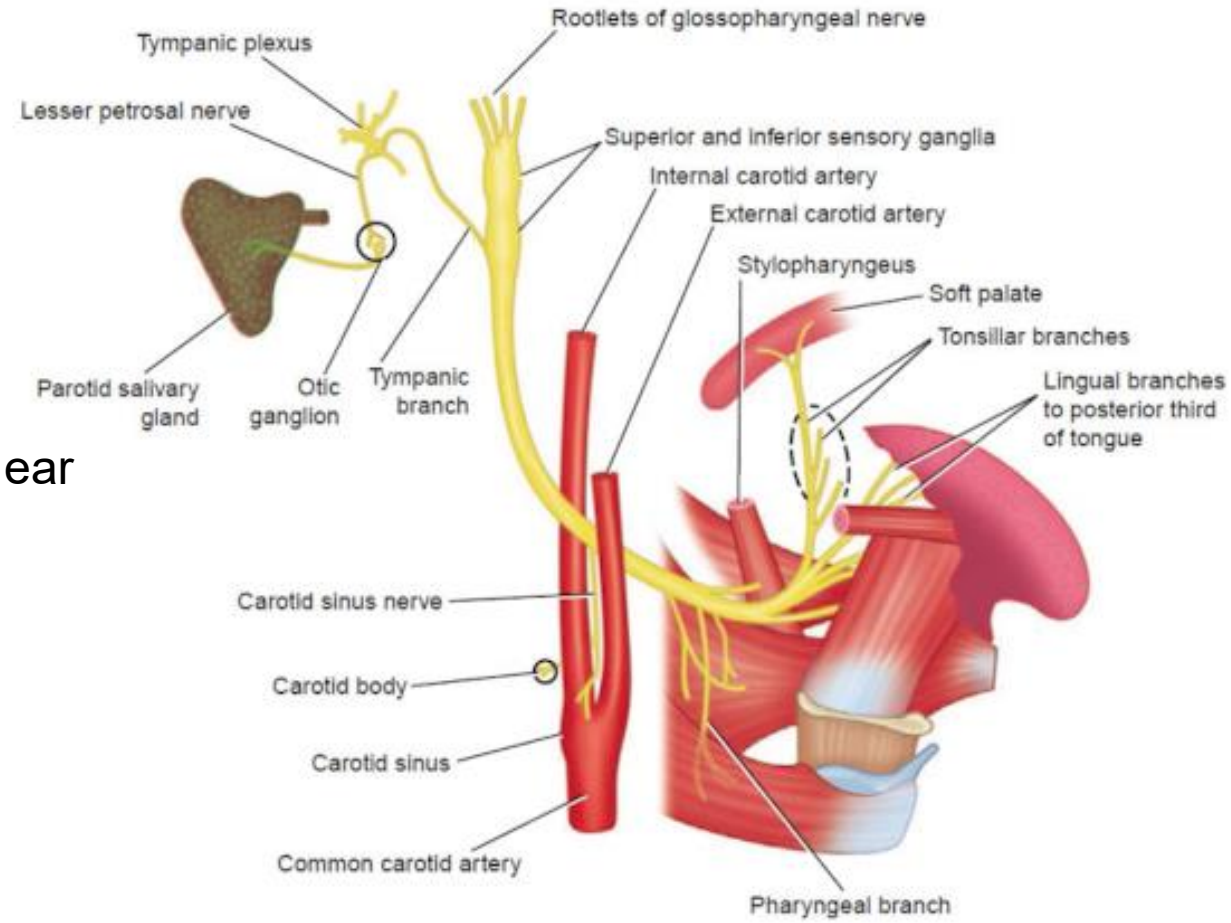


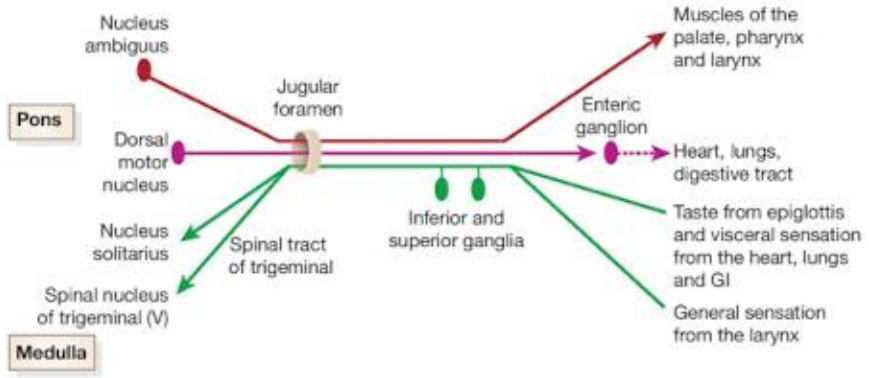
Fig. 11.17. Distribution of the glossopharyngeal nerve.

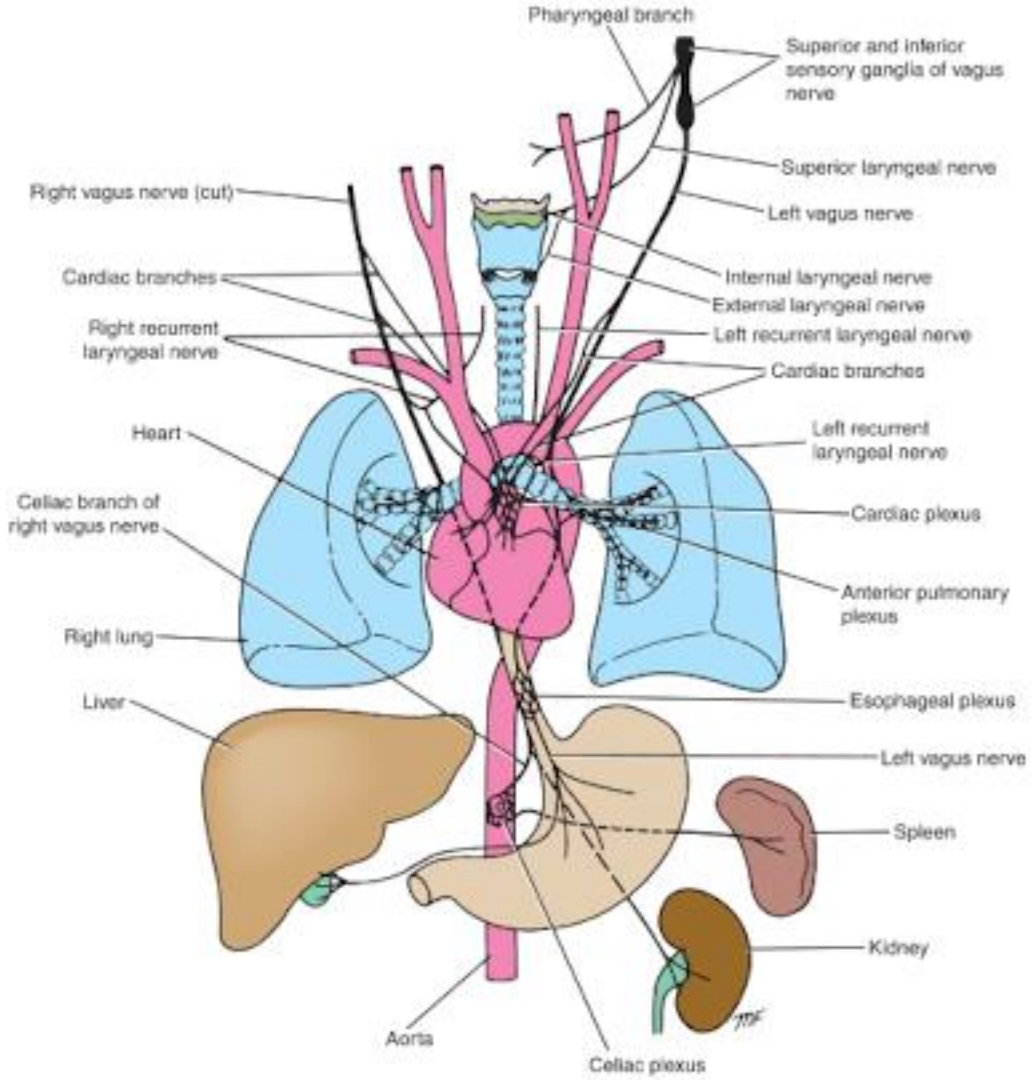
CN IX = posterior 1/3 tongue + carotid sinus/body + stylopharyngeus + parotid

Images: Champney; Splittgerber, SNA, 8e

Vagus Nerve (CN X)

- Origin: medulla
- **SVE:** most muscles of soft palate, pharynx & larynx
- **GVE:** parasympathetic to thoracic & abdominal viscera
- **SVA:** taste from epiglottis/posterior tongue
- **GVA:** thoracic & abdominal viscera
- **GSA:** pharynx, larynx & external ear
- Exits skull through jugular foramen





CN X = swallowing + phonation + visceral parasympathetic

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Fig. 12.65. Distribution of the vagus nerve.
Wineski, SCABR, 10e.

CN IX & X: Key Brainstem Nuclei

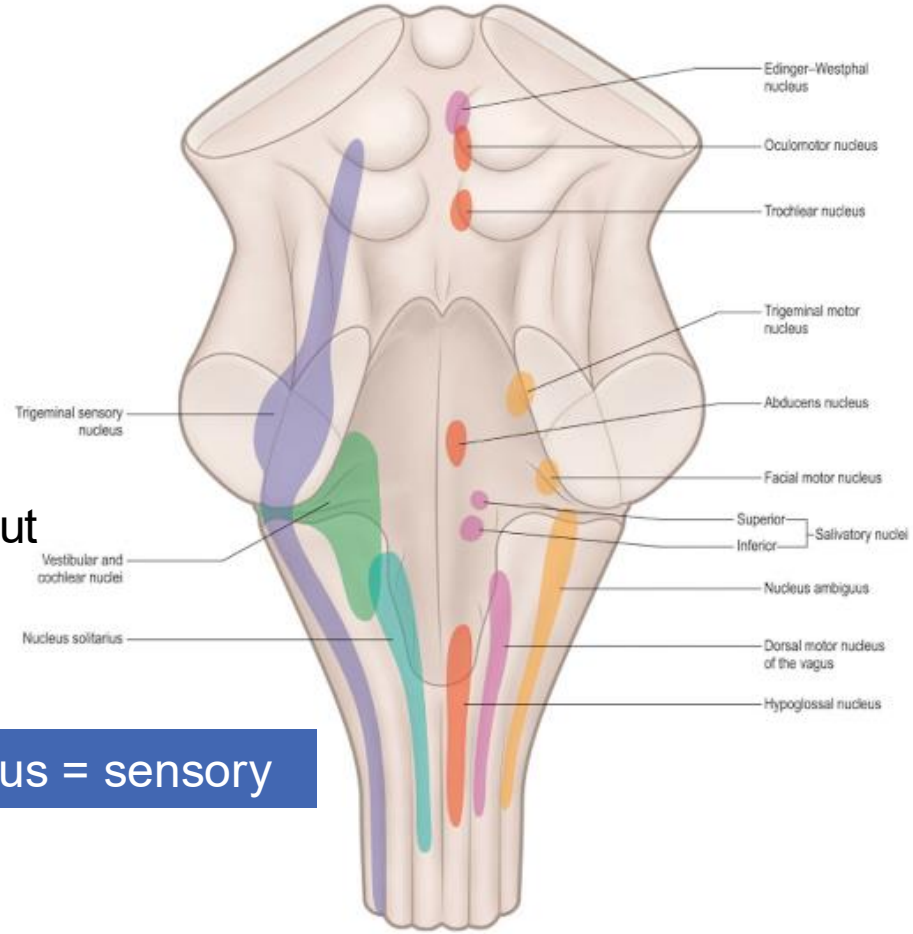
Motor (Efferent)

- Nucleus ambiguus (SVE) → pharyngeal/laryngeal motor
- Inferior salivatory nucleus (GVE) → CN IX → parotid gland
- Dorsal motor nucleus of X (GVE) → thoracoabdominal viscera

Sensory (Afferent)

- Nucleus solitarius (GVA & SVA) → taste & visceral sensory input
- Spinal trigeminal nucleus (GSA) → somatic sensation

Ambiguus = motor | Dorsal motor X = parasympathetic | Solitarius = sensory



Dorsal brainstem view of the cranial nerve nuclei. Crossman, 7e.

Vagus Motor Function & Lesion

- Weakness of palate, pharynx & larynx
- Dysphagia
- Hoarseness
- Impaired efferent limb of gag reflex
- Palate droops on affected side
- Uvula deviates away from the lesion

CN X lesion → palate droops ipsilaterally; uvula points away



Singh. *Neurosciences,* 2015.

Pharyngeal (Gag) Reflex

- Touch posterior pharynx → bilateral pharyngeal contraction
- **Afferent limb:** CN IX
- **Efferent limb:** CN X
- Motor response → pharyngeal elevation/contraction

Gag reflex: IX in → X out

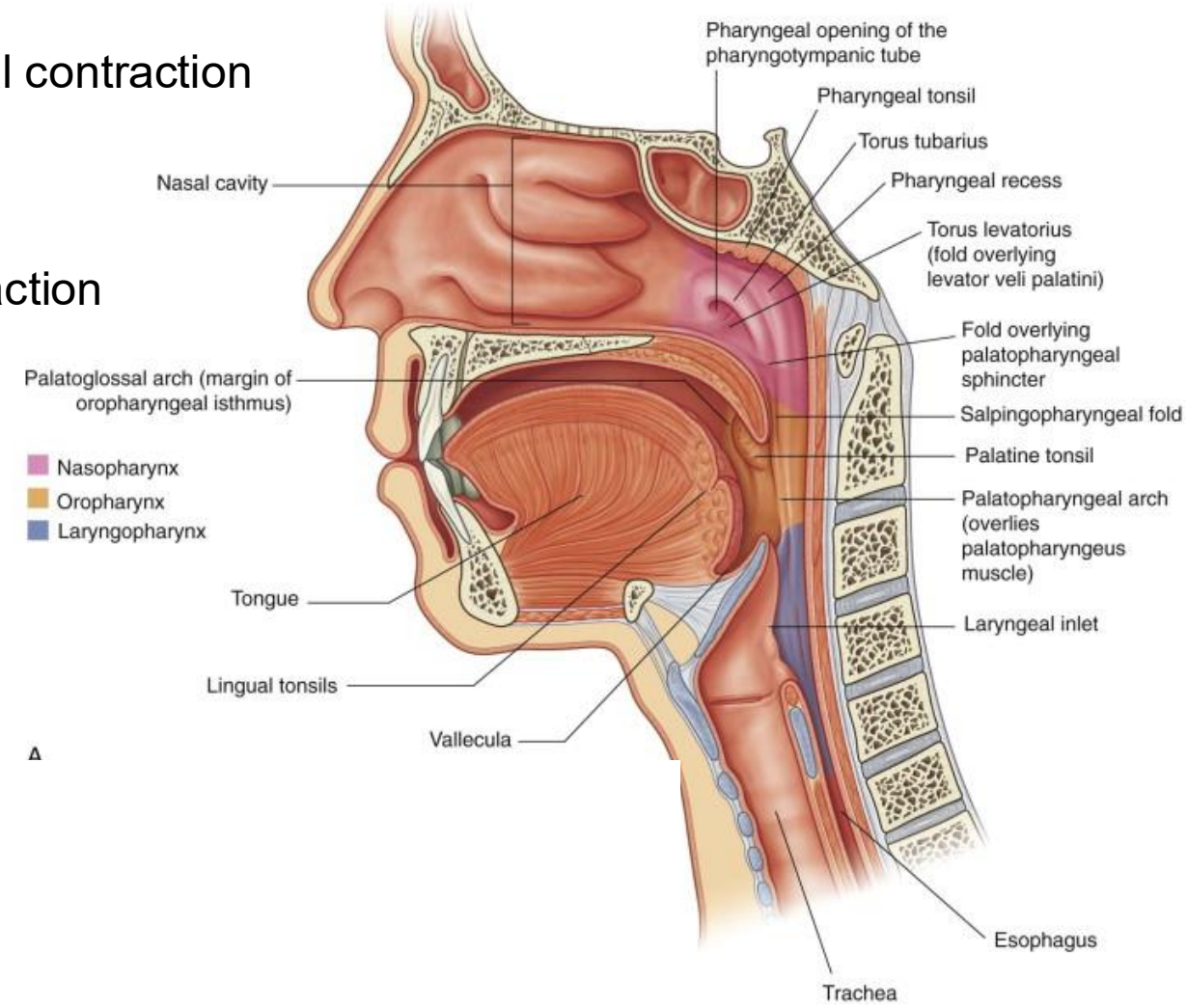


Fig. 8.205 GAS, 4e.

Larynx Innervation (CN X)

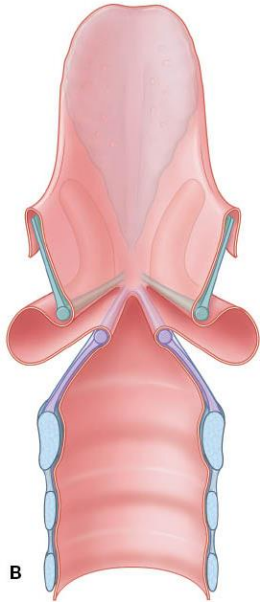
Superior Laryngeal Nerve

- External laryngeal nerve: motor → cricothyroid
- Internal laryngeal nerve: sensory → above cords

Inferior (Recurrent) Laryngeal Nerve

- Motor → all laryngeal muscles except cricothyroid
- Sensory → below vocal folds

Right RLN → loops under right SCA
Left RLN → loops under aortic arch



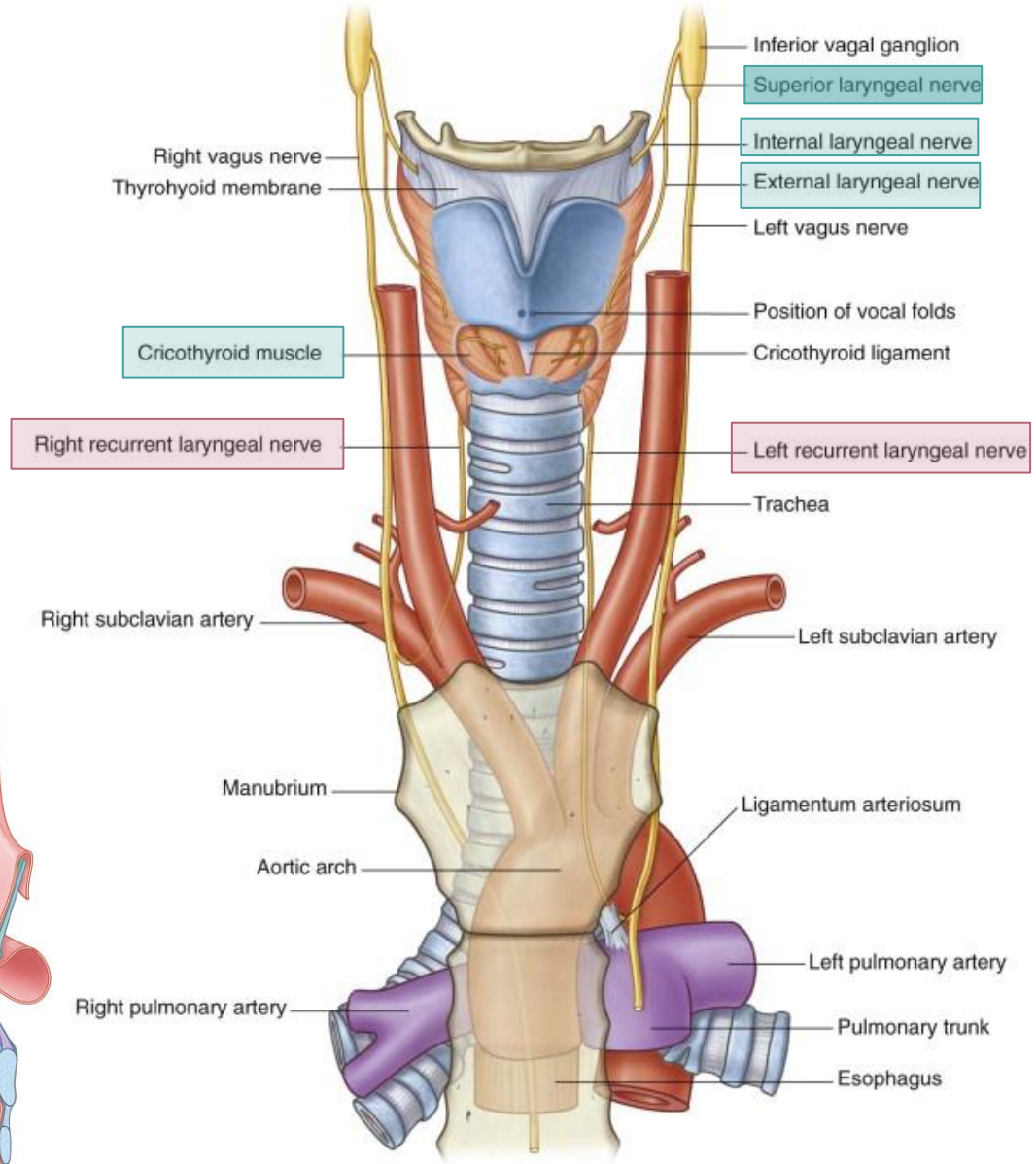


Fig. 8.221, 8.228 Gray's Anatomy for Students, 4e.

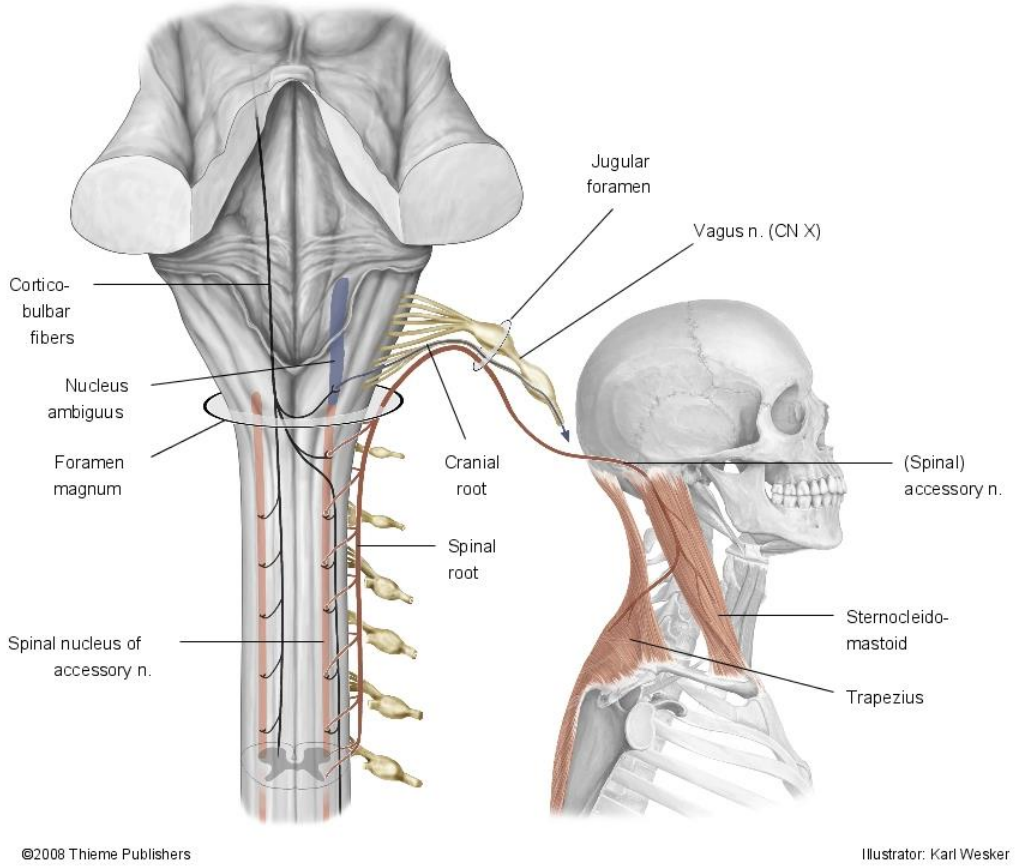
Head, Neck & Tongue Motor Function

CN XI • CN XII • Head & Shoulder Movement • Tongue Movement

Accessory Nerve (CN XI)

- Origin: spinal cord C1–C5
- Motor → sternocleidomastoid (SCM) + trapezius
- Spinal roots ascend through foramen magnum
- Exits skull through jugular foramen
- SCM → rotates head to opposite side
- Trapezius → elevates shoulder



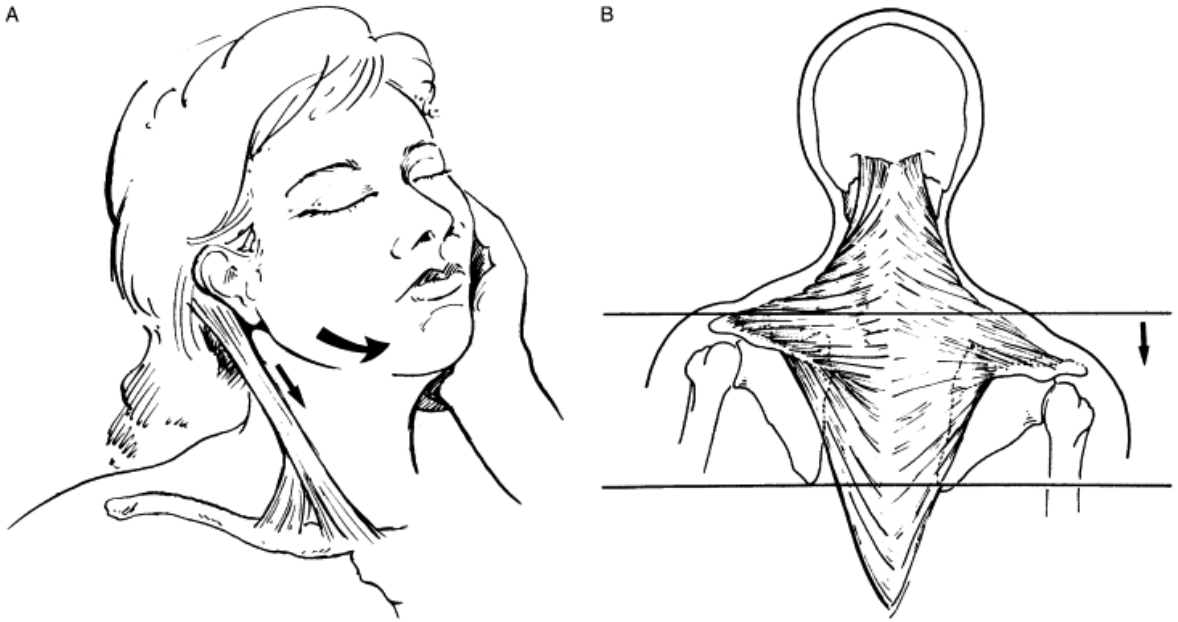


CN XI → SCM + trapezius → head rotation + shoulder elevation

Champney. ECNA, 2016. Thieme

Accessory Nerve Lesion

- Ipsilateral SCM weakness
- Difficulty turning head away from the lesion
- Ipsilateral trapezius weakness
- Weak shoulder shrug on affected side
- May produce shoulder droop

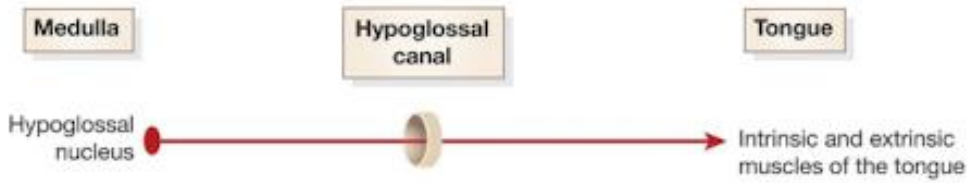


Addas, Clarke. *Encyclopedia of the Neurological Sciences*. 2003.

CN XI lesion → weak ipsilateral shoulder shrug + difficulty turning head contralaterally

Hypoglossal Nerve (CN XII)

- Origin: medulla
- General somatic efferent (GSE)
- Motor to intrinsic + extrinsic tongue muscles
- Exception: palatoglossus → CN X
- Exits skull through hypoglossal canal
- Genioglossus protrudes the tongue



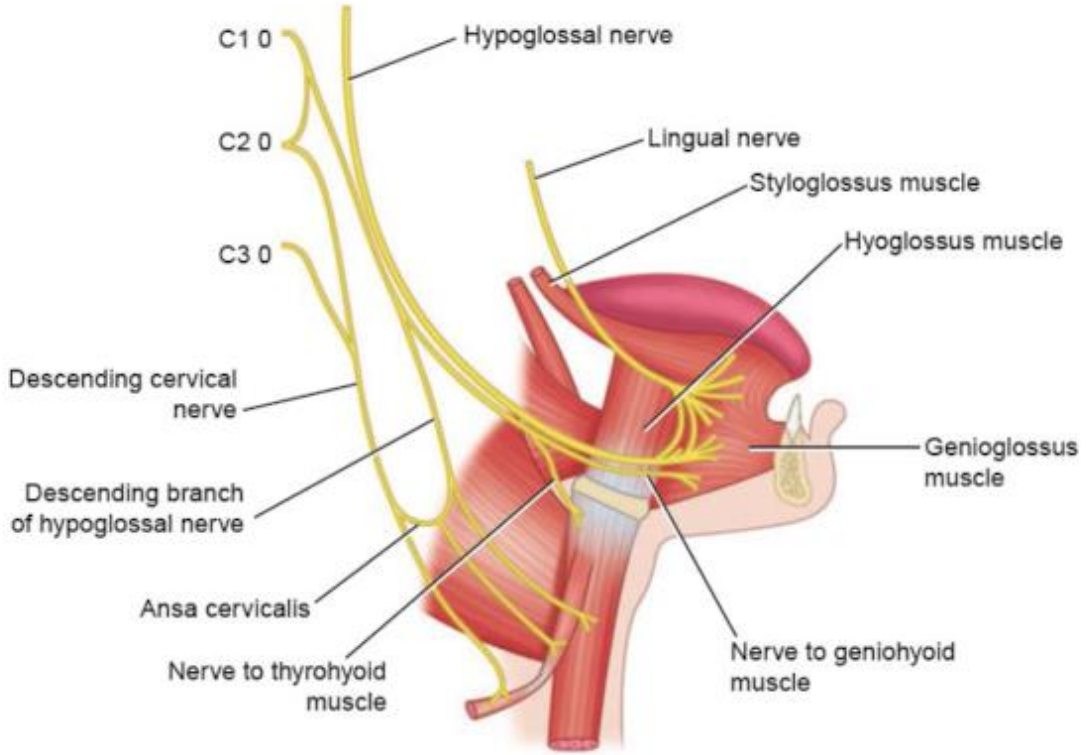


Figure 11-23 Distribution of the hypoglossal nerve.

CN XII = motor to tongue | Palatoglossus = CN X

Singh. TBCNA, 3e.
Champney. ECNA, 2016.

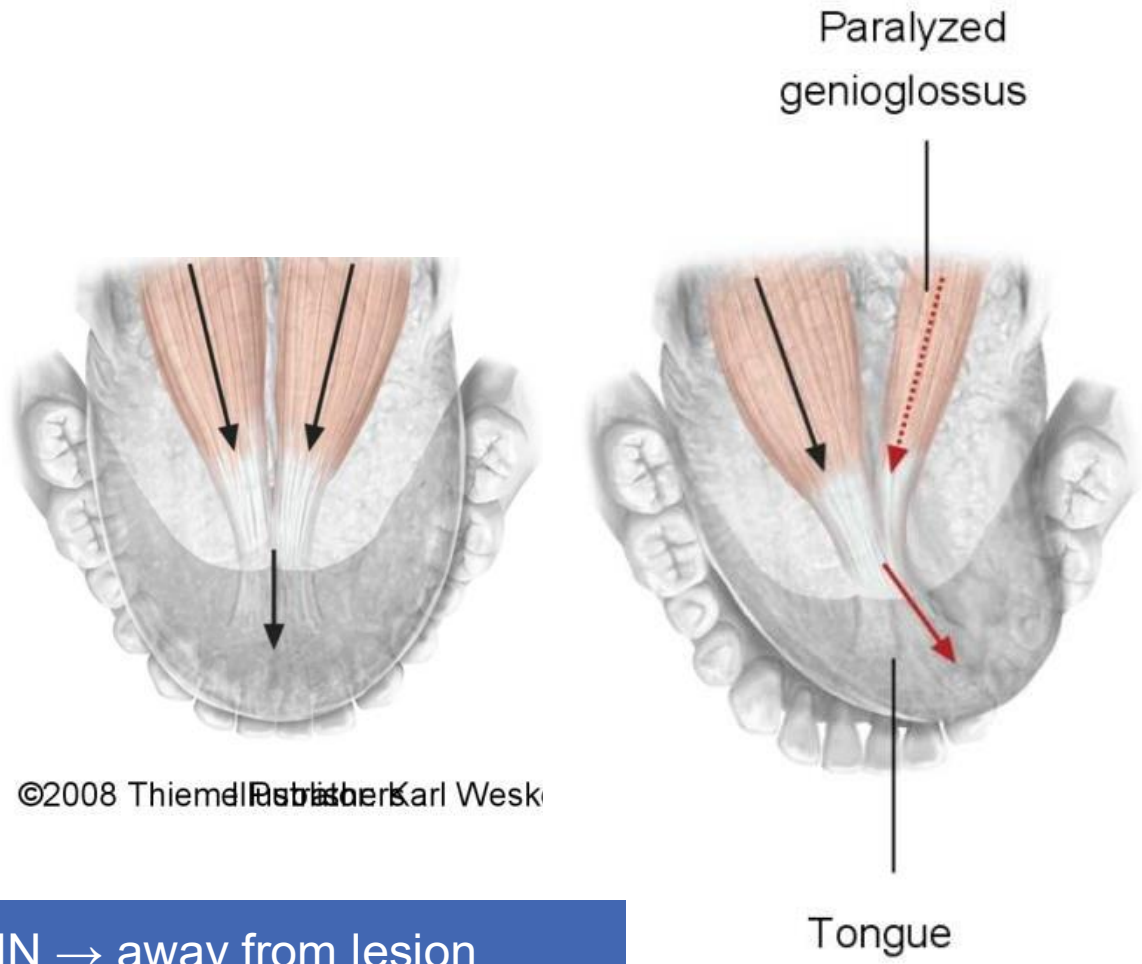
Hypoglossal Lesions: UMN vs LMN

LMN / CN XII Lesion

- Ipsilateral tongue weakness
- Atrophy + fasciculations
- Tongue deviates toward the lesion on protrusion

UMN / Corticobulbar Lesion

- Contralateral genioglossus weakness
- No prominent atrophy/fasciculations
- Tongue deviates away from the cortical lesion



©2008 Thieme | Illustrations: Karl Wesk

LMN → toward lesion | UMN → away from lesion

The tongue deviates toward the weak genioglossus

Tongue Innervation

Anterior 2/3

- General sensation (GSA): lingual n. (CN V3)
- Taste (SVA): chorda tympani (CN VII)

Posterior 1/3

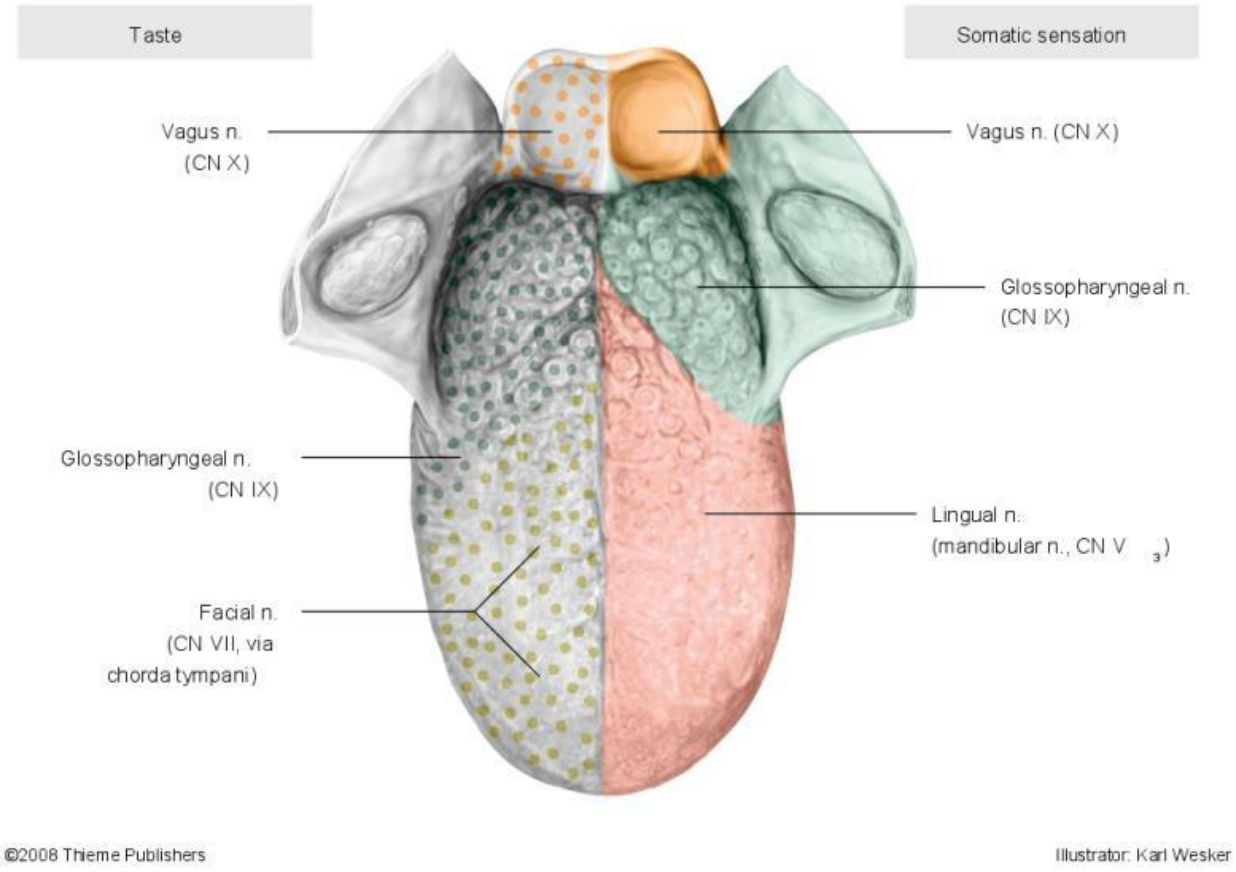
- General sensation (GSA): CN IX
- Taste (SVA): CN IX

Most Posterior Tongue & Epiglottis

- General sensation + taste: CN X

Motor

- CN XII: intrinsic + extrinsic tongue muscles
- Exception: palatoglossus → CN X



Internal Organization of the Brainstem

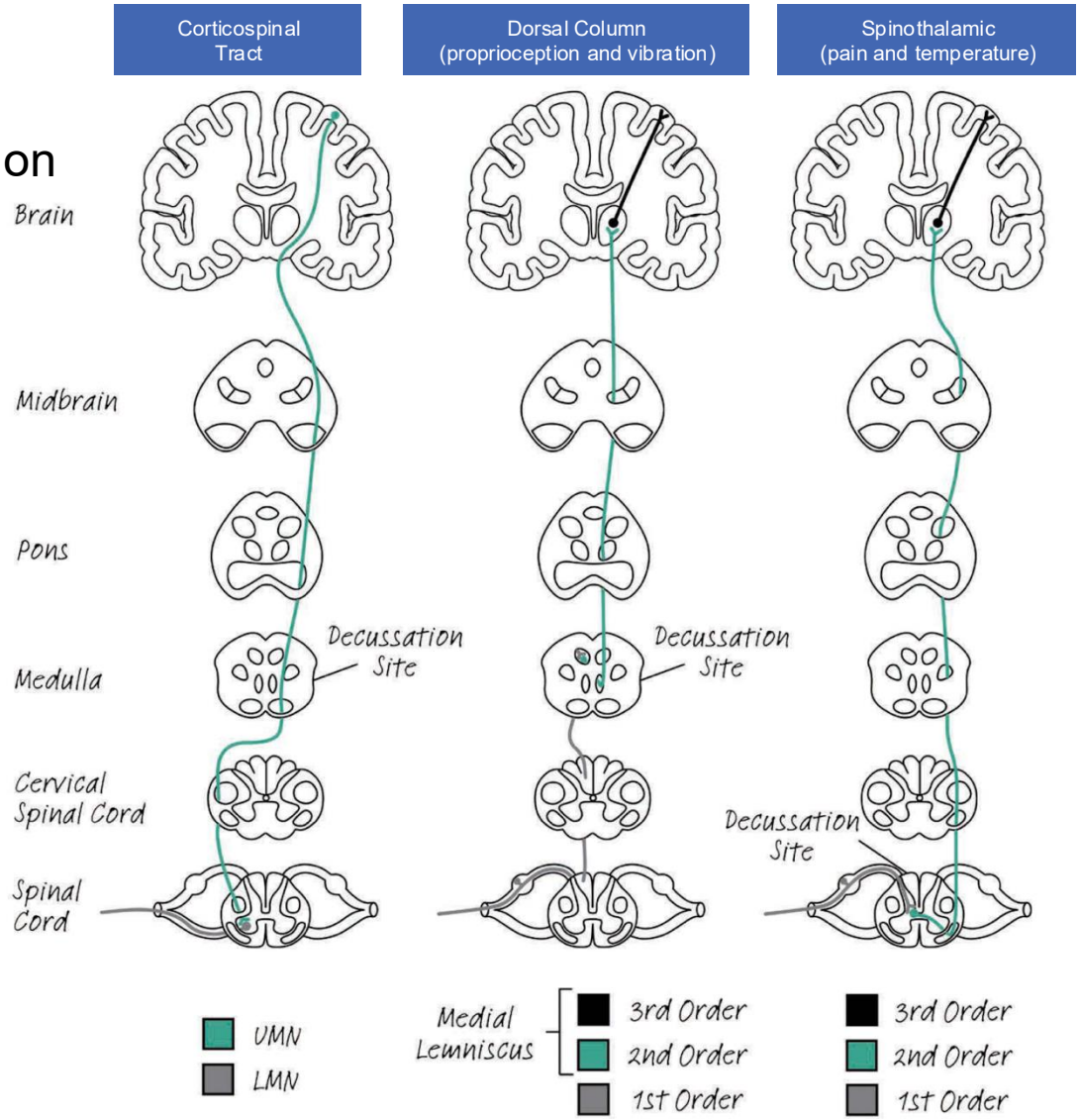
Long Tracts • Cross-Sections • Vascular Territories

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Major Long Tracts Through the Brainstem

- **Corticospinal tract** → voluntary motor control
- **Medial lemniscus** → fine touch, vibration, proprioception
- **Spinothalamic tract** → pain & temperature

In the brainstem, these pathways have already crossed or will cross at predictable levels.



LMN: lower motor neuron
UMN: upper motor neuron

WB, NA

Long Tracts in the Brainstem

Corticospinal Tract

- Crosses in caudal medulla
- Brainstem lesion above crossing → contralateral weakness

Medial Lemniscus

- Crosses in caudal medulla
- Brainstem lesion above crossing → contralateral loss of fine touch, vibration & proprioception

Spinothalamic Tract

- Already crossed in spinal cord
- Brainstem lesion → contralateral loss of pain & temperature

Face vs Body Sensory Pathways

Face

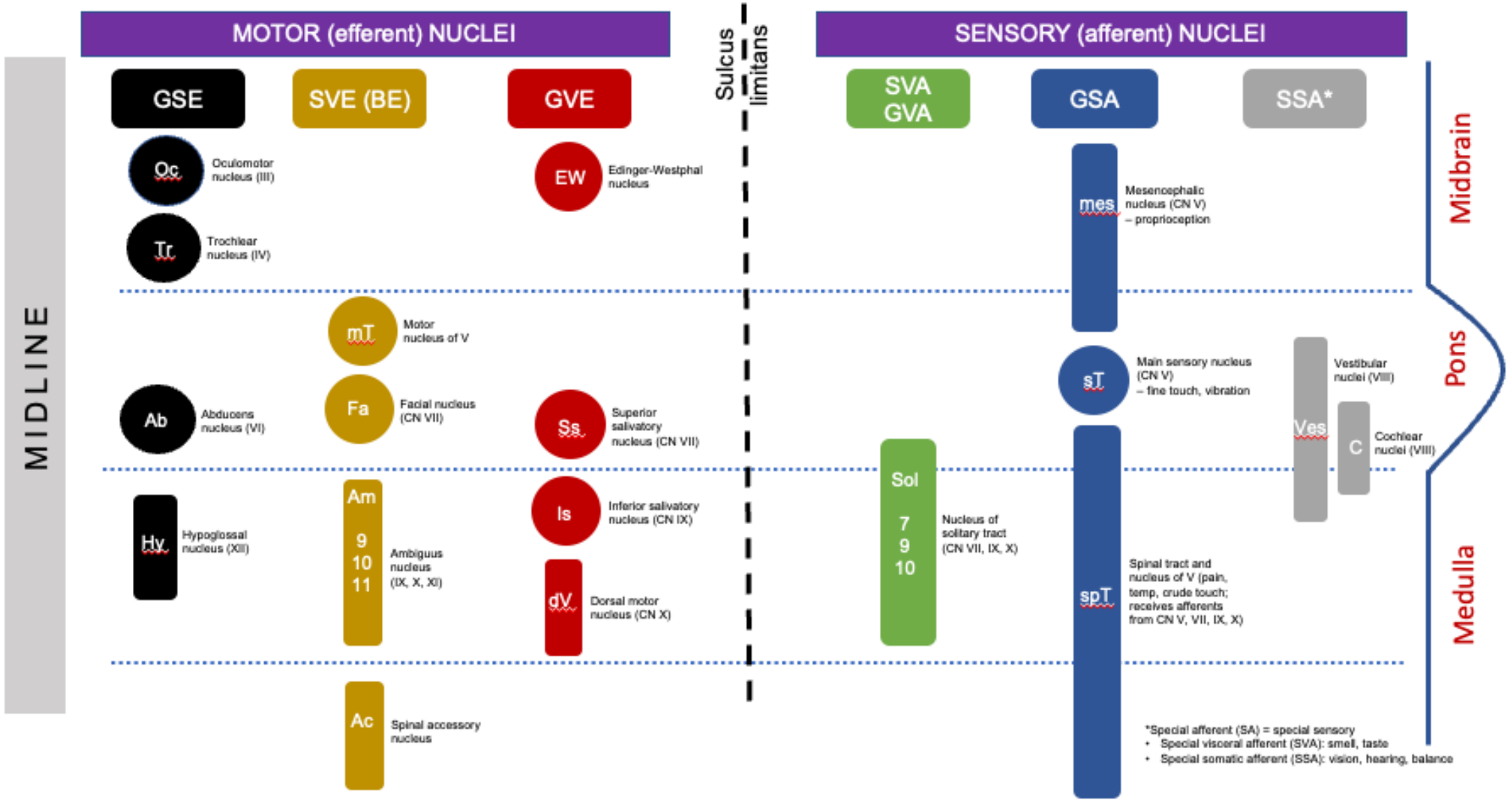
- Principal sensory nucleus of V → fine touch/vibration
- Spinal trigeminal nucleus/tract → pain & temperature

Body

- Medial lemniscus → fine touch/vibration/proprioception
- Spinothalamic tract → pain & temperature

Brainstem lesion may produce ipsilateral face + contralateral body deficits

Cranial Nerve Nuclei in the Brainstem



How to Read a Brainstem Cross-Section

Identify the level

- Midbrain | pons | medulla

Find orientation landmarks:

- Aqueduct | 4th ventricle | colliculi | olive/pyramids

Identify CN nuclei/fascicles:

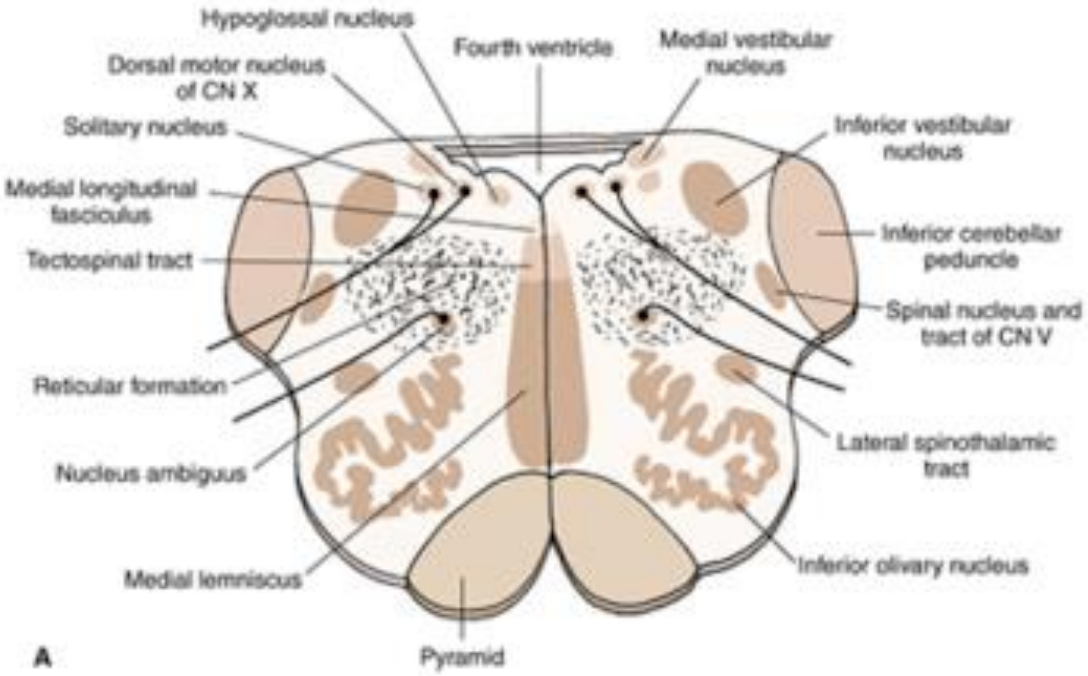
- Which cranial nerves belong at this level?

Locate major long tracts:

- Corticospinal | medial lemniscus | spinothalamic

Orient medial vs lateral:

- Prepare to localize lesions



Cross-Section of Central Medulla

Landmark → CN structures → Long tracts → Medial/Lateral

Fig. 9.8. Siegel, EN, 4e.

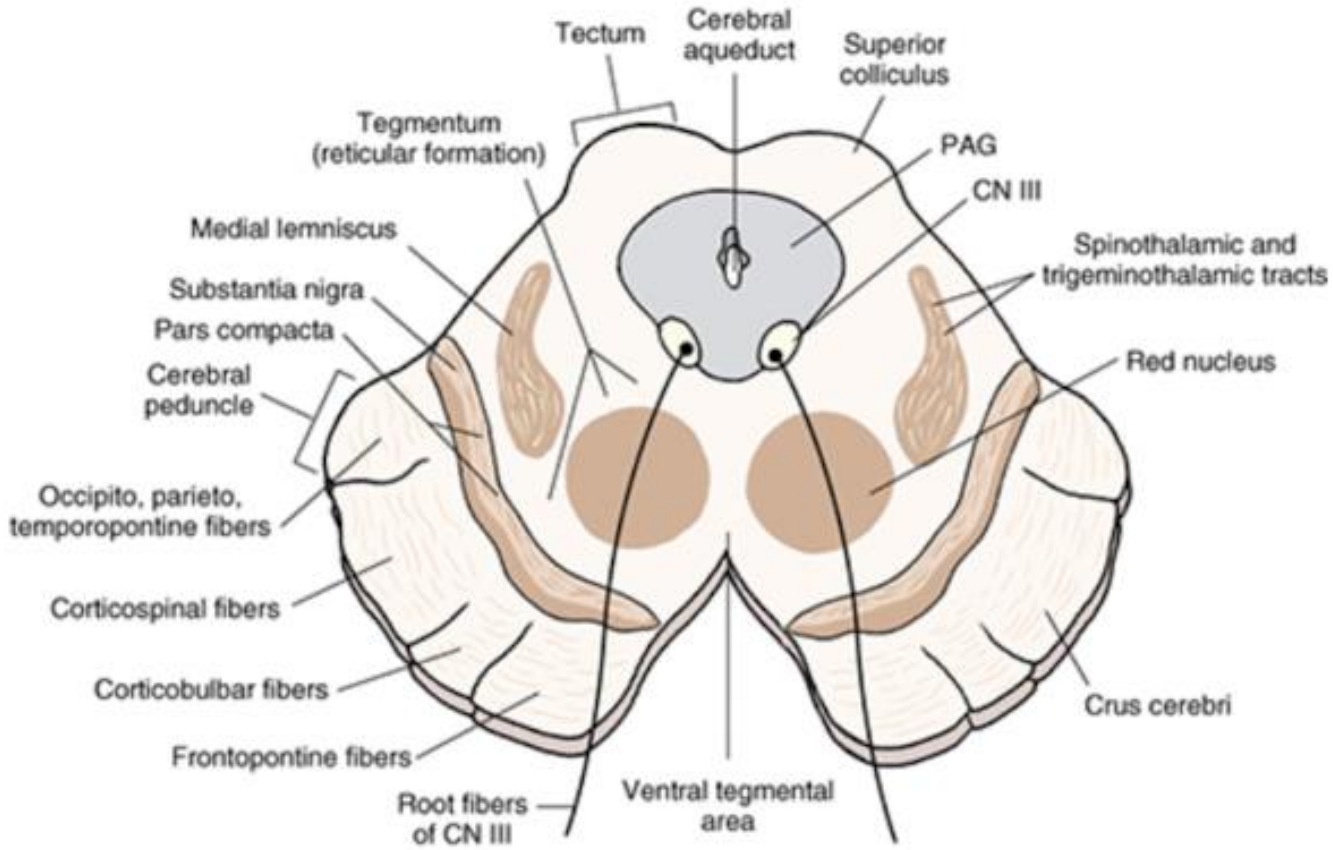
Midbrain: Superior Colliculus Level

Landmarks
- Superior colliculus → visual reflex center
- Cerebral aqueduct

CN Structure
- Oculomotor nucleus (CN III)

Long Tracts
- Cerebral peduncle (crus cerebri) → corticospinal fibers
- Medial lemniscus
- Spinothalamic tract

Other Key Structures
- Red nucleus
- Substantia nigra



Superior colliculus + CN III → rostral midbrain

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Fig. 11.5. Siegel, EN, 4e.

Midbrain: Inferior Colliculus Level

Landmarks

- Inferior colliculus → auditory reflex center
- Cerebral aqueduct

CN Structure

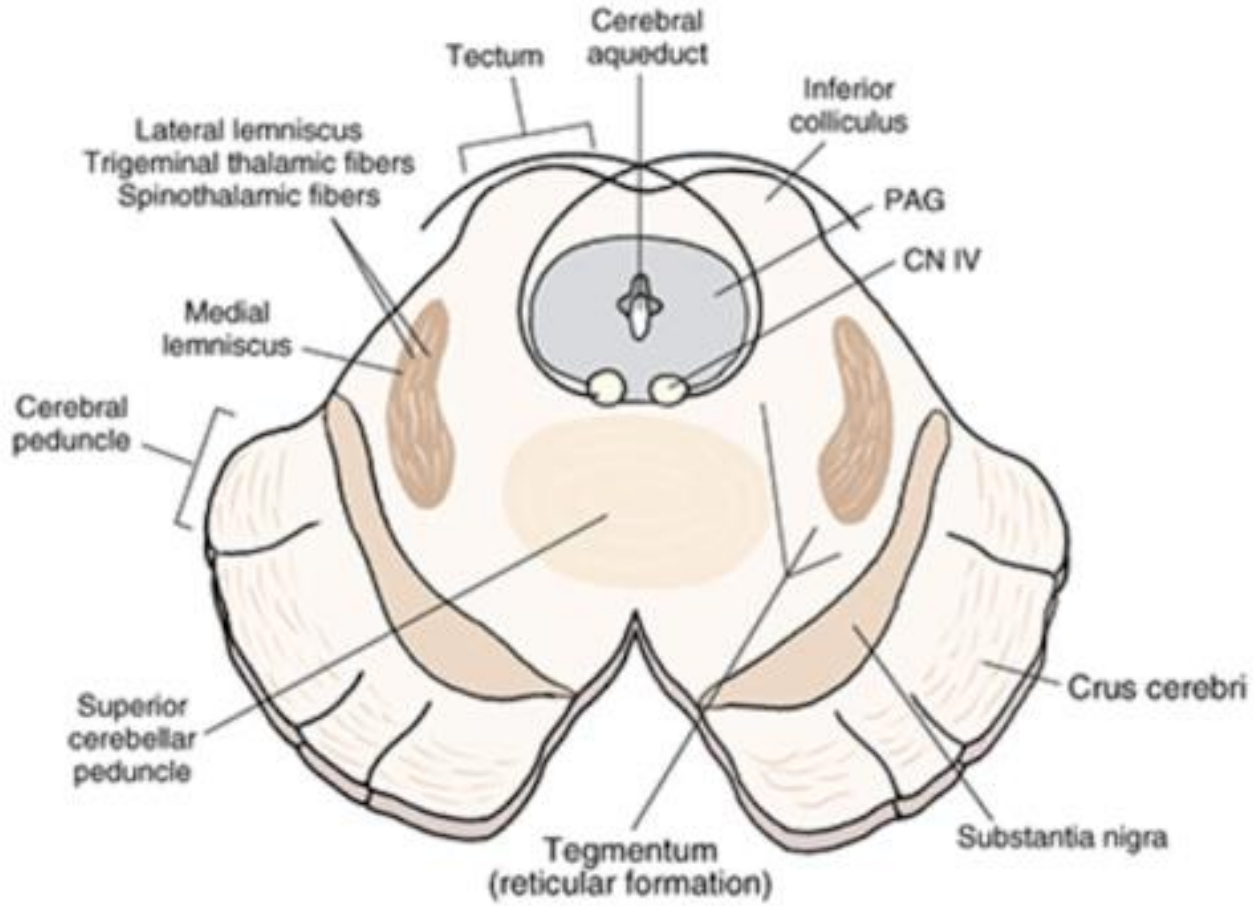
- Trochlear nucleus (CN IV)

Long Tracts

- Cerebral peduncle → corticospinal fibers
- Medial lemniscus
- Spinothalamic tract

Other Key Structure

- Superior cerebellar peduncle decussation
- Substantia nigra



Inferior colliculus + CN IV → caudal midbrain

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Fig. 11.3. Siegel, EN, 4e.

Rostral Pons

Landmarks

- Large basis pontis
- Middle cerebellar peduncle

CN Structure

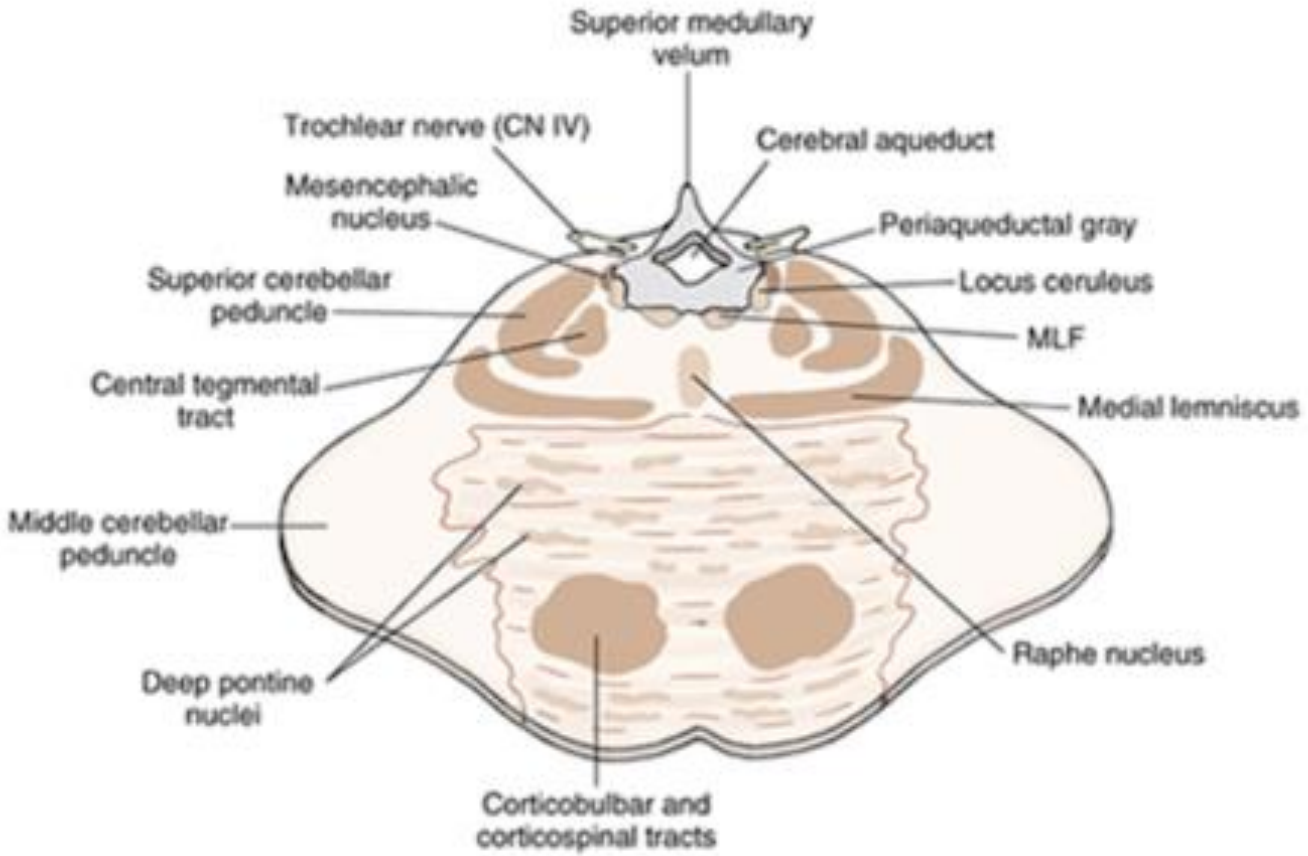
- Mesencephalic nucleus of CN V

Long Tracts

- Corticospinal tract
- Medial lemniscus

Other Key Structures

- Pontine nuclei



Large basis pontis + middle cerebellar peduncle → pons

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV

Fig. 10.6. Siegel, EN, 4e.

Middle Pons

Landmarks

- 4th ventricle
- Middle cerebellar peduncle

CN Nuclei

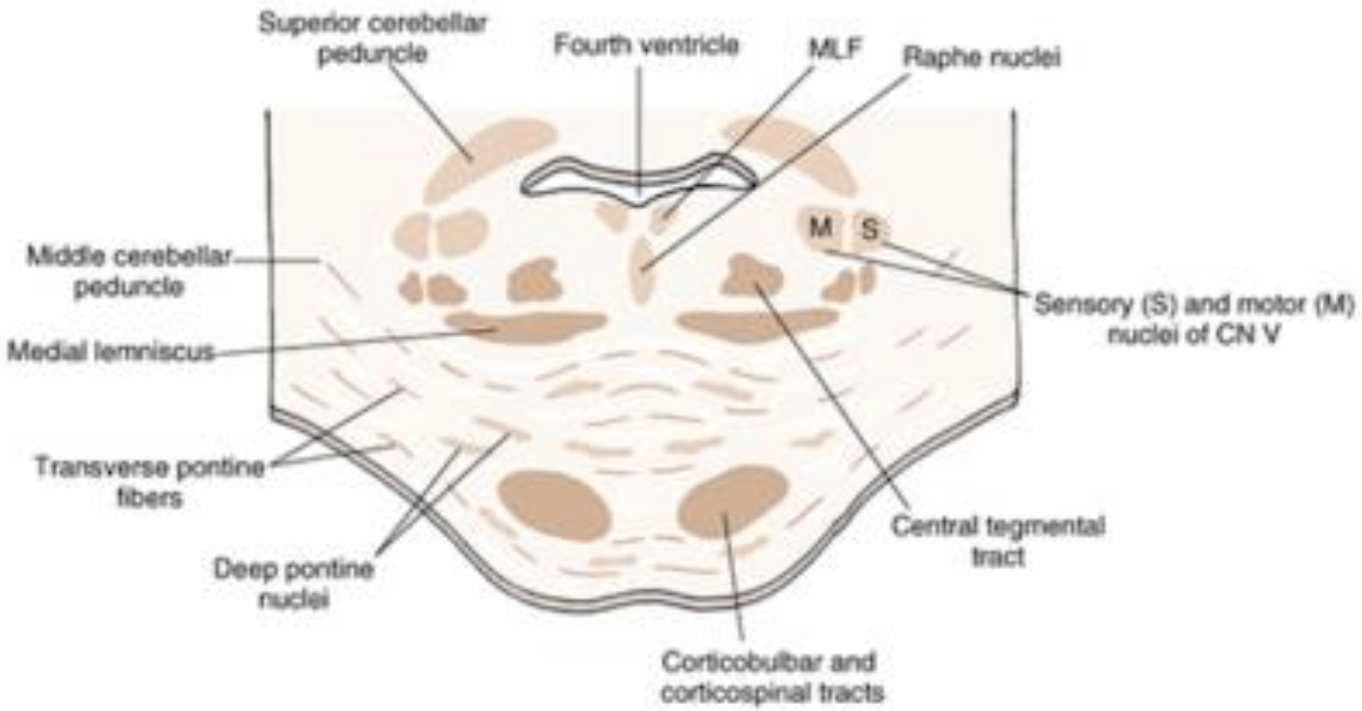
- Principal sensory nucleus of CN V
- Motor nucleus of CN V

Long Tracts

- Corticospinal tract
- Medial lemniscus
- Medial longitudinal fasciculus (MLF)

Other Key Structures

- Pontine nuclei + transverse pontine fibers



CN V nuclei → mid-pons

Fig. 10.5. Siegel, EN, 4e.

Caudal Pons

Landmark

- Facial colliculus
- 4th ventricle

CN Nuclei

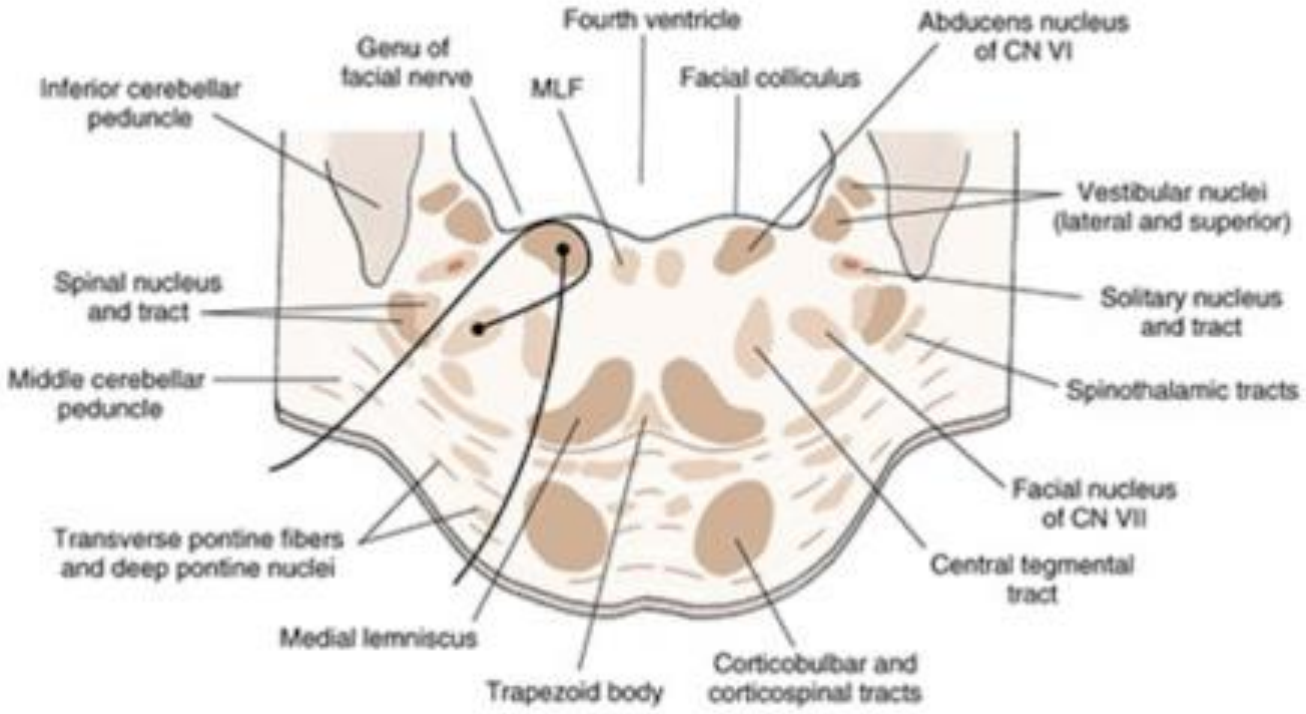
- Abducens nucleus (CN VI)
- Facial nucleus (CN VII)
- Vestibular nuclei (CN VIII)
- Spinal trigeminal nucleus & tract

Long Tracts

- Corticospinal tract
- Medial lemniscus

Cerebellar Connections

- Middle + inferior cerebellar peduncles



CN VI/VII + facial colliculus → caudal pons

Fig. 10.4. Siegel, EN, 4e.

Upper (Open) Medulla

Landmarks

- 4th ventricle
- Inferior olive

CN/Sensory Nuclei

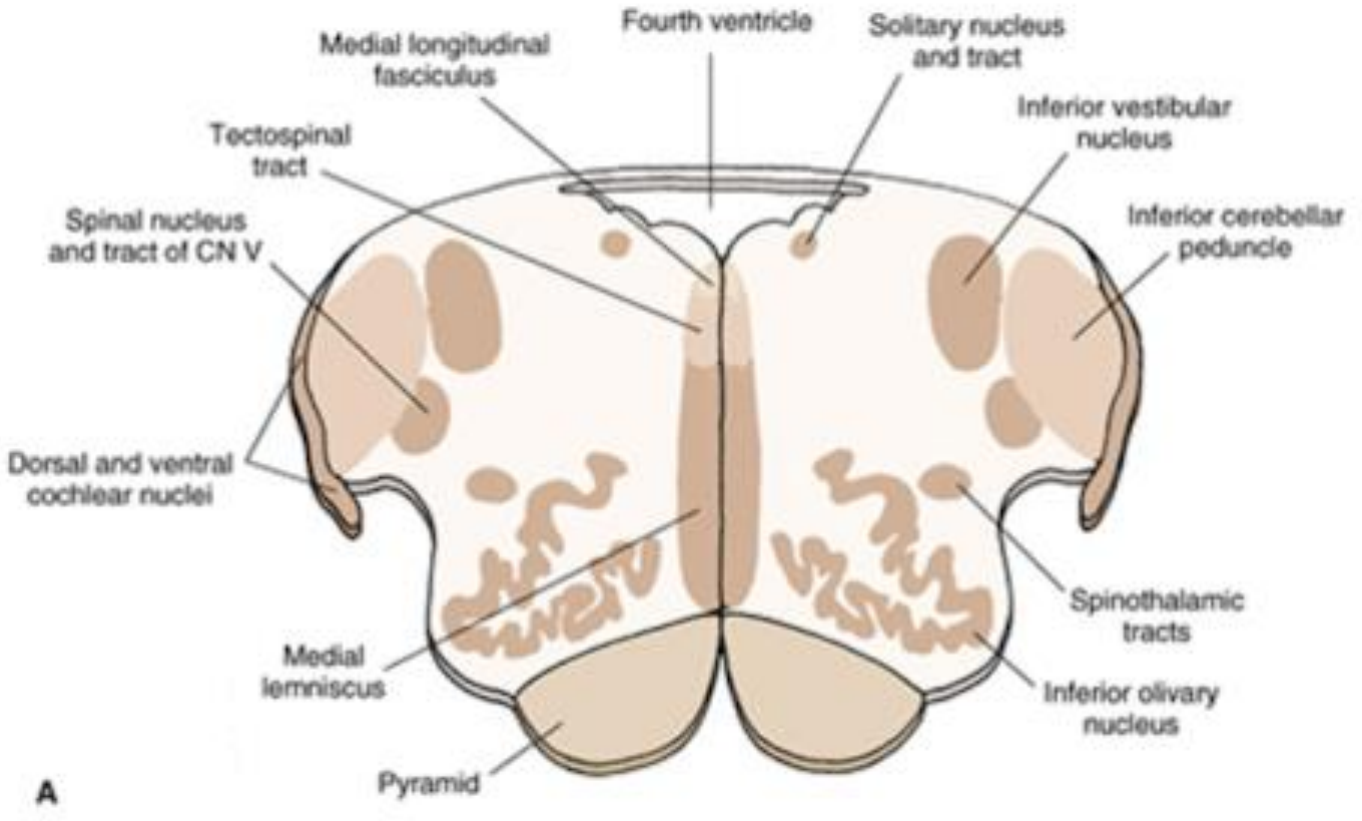
- Vestibular nuclei
- Solitary nucleus
- Spinal trigeminal nucleus & tract

Long Tracts

- Pyramid → corticospinal tract
- Medial lemniscus

Cerebellar Connection

- Inferior cerebellar peduncle



Olive + open 4th ventricle → rostral/open medulla

Fig. 9.8. Siegel, EN, 4e.

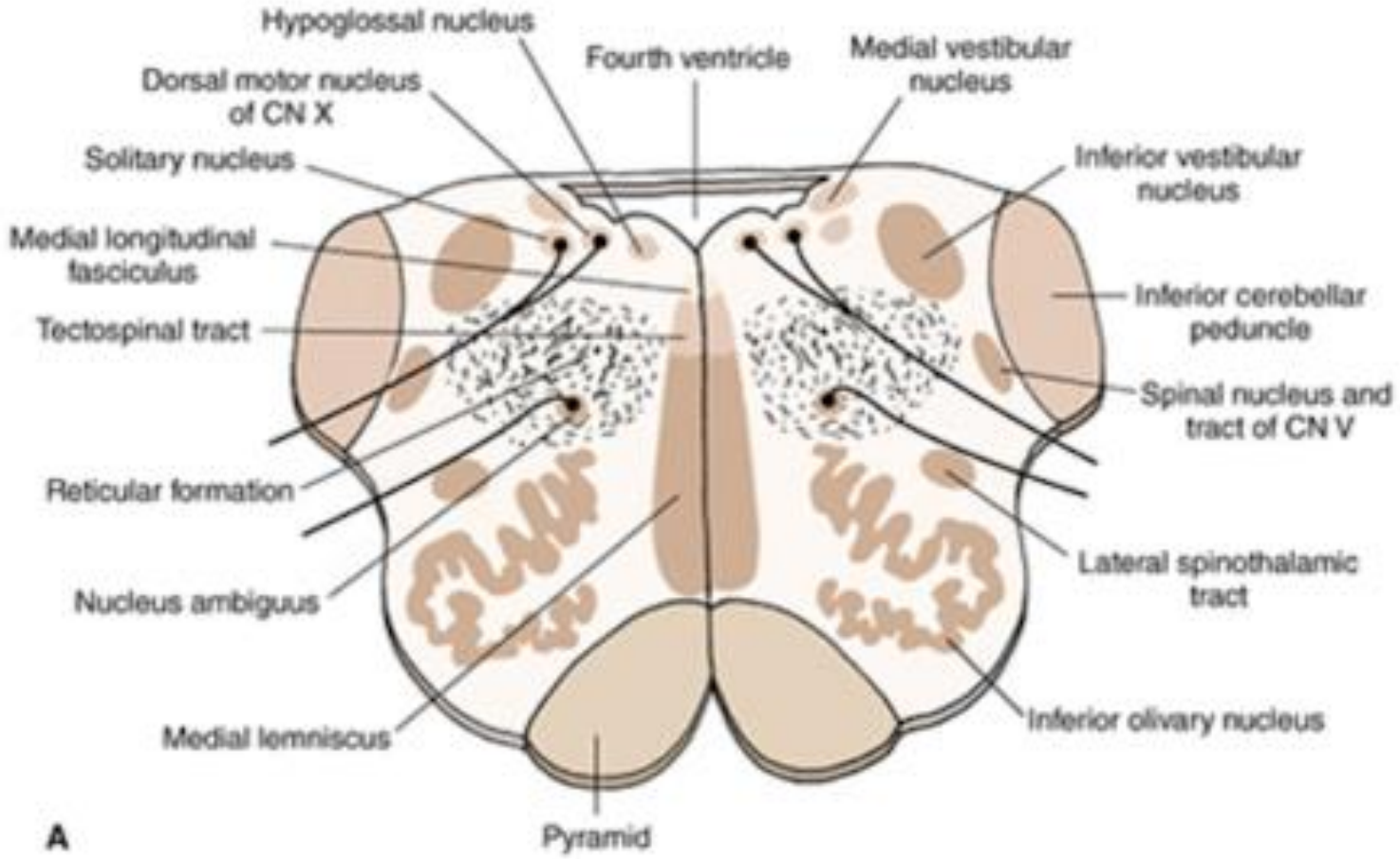
Central Medulla

Landmarks
- 4th ventricle
- Inferior olive

CN Nuclei
- Hypoglossal nucleus (CN XII)
- Dorsal motor nucleus of X
- Nucleus ambiguus
- Solitary nucleus
- Spinal trigeminal nucleus & tract

Long Tracts
- Pyramid → corticospinal tract
- Medial lemniscus
- Spinothalamic tract

Cerebellar Connection
- Inferior cerebellar peduncle



Medial structures → XII + pyramid + medial lemniscus
Lateral structures → ambiguus + spinal V + spinothalamic

Fig. 9.8. Siegel, EN, 4e.

Lower (Closed) Medulla

Landmark

- Central canal

Sensory Structures

- Gracile + cuneate nuclei
- Internal arcuate fibers → sensory decussation
- Formation of medial lemniscus

Motor Structure

- Pyramids
- Pyramidal decussation

Face/Body Pain Pathways

- Spinal trigeminal nucleus & tract
- Spinothalamic tract

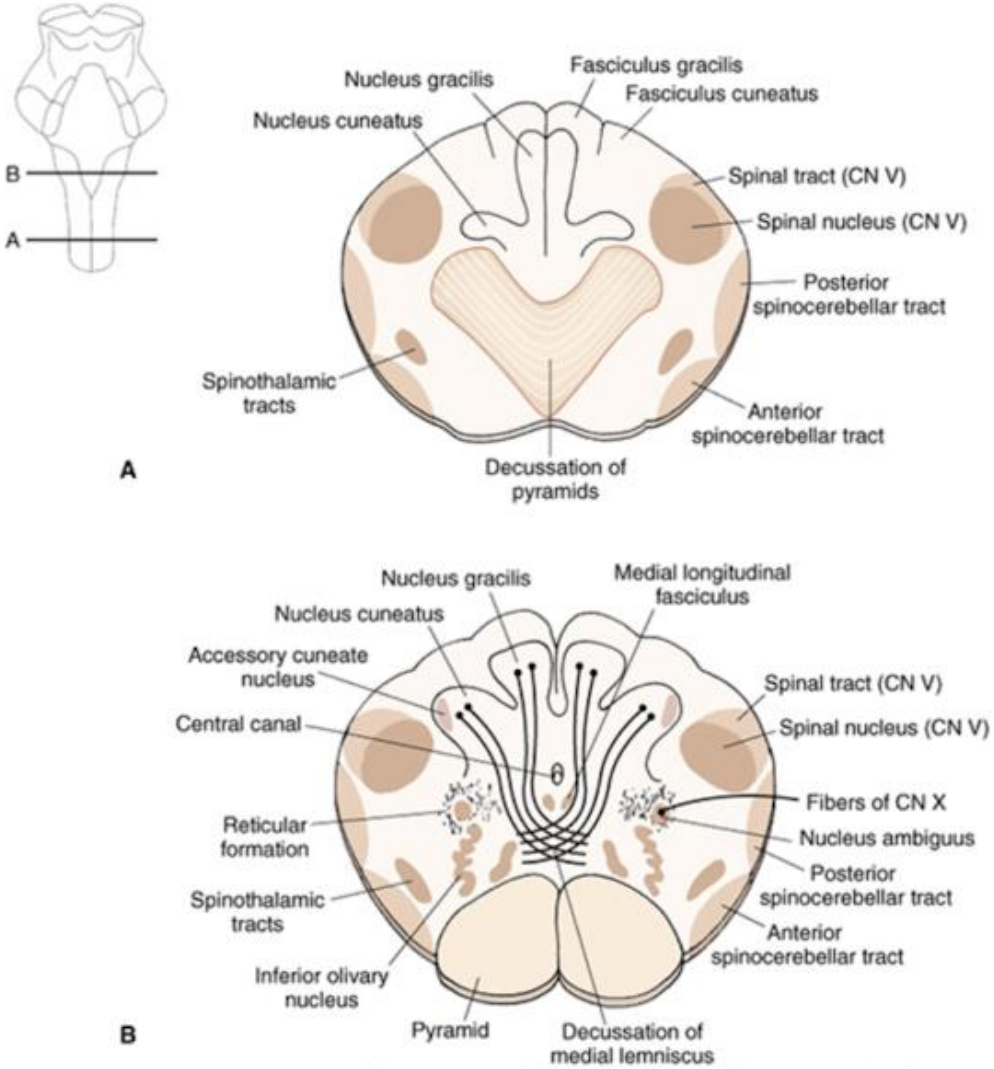


FIGURE 9.3 Cross-sectional diagrams of the lower medulla. **A.** Level of the decussation of the pyramids. **B.** Level of the decussation of the medial lemniscus. *CN*, cranial nerve.

Caudal medulla = two major crossings: sensory decussation + pyramidal decussation

Fig. 9.3. Siegel, EN, 4e.

Brainstem Vascular Territories

Midbrain

- Posterior cerebral artery (PCA)
- Superior cerebellar artery (SCA)

Pons

- Pontine branches of basilar artery
- Anterior inferior cerebellar artery (AICA)
- Superior cerebellar artery (SCA)

Medulla

- Vertebral artery
- Posterior inferior cerebellar artery (PICA)
- Anterior spinal artery

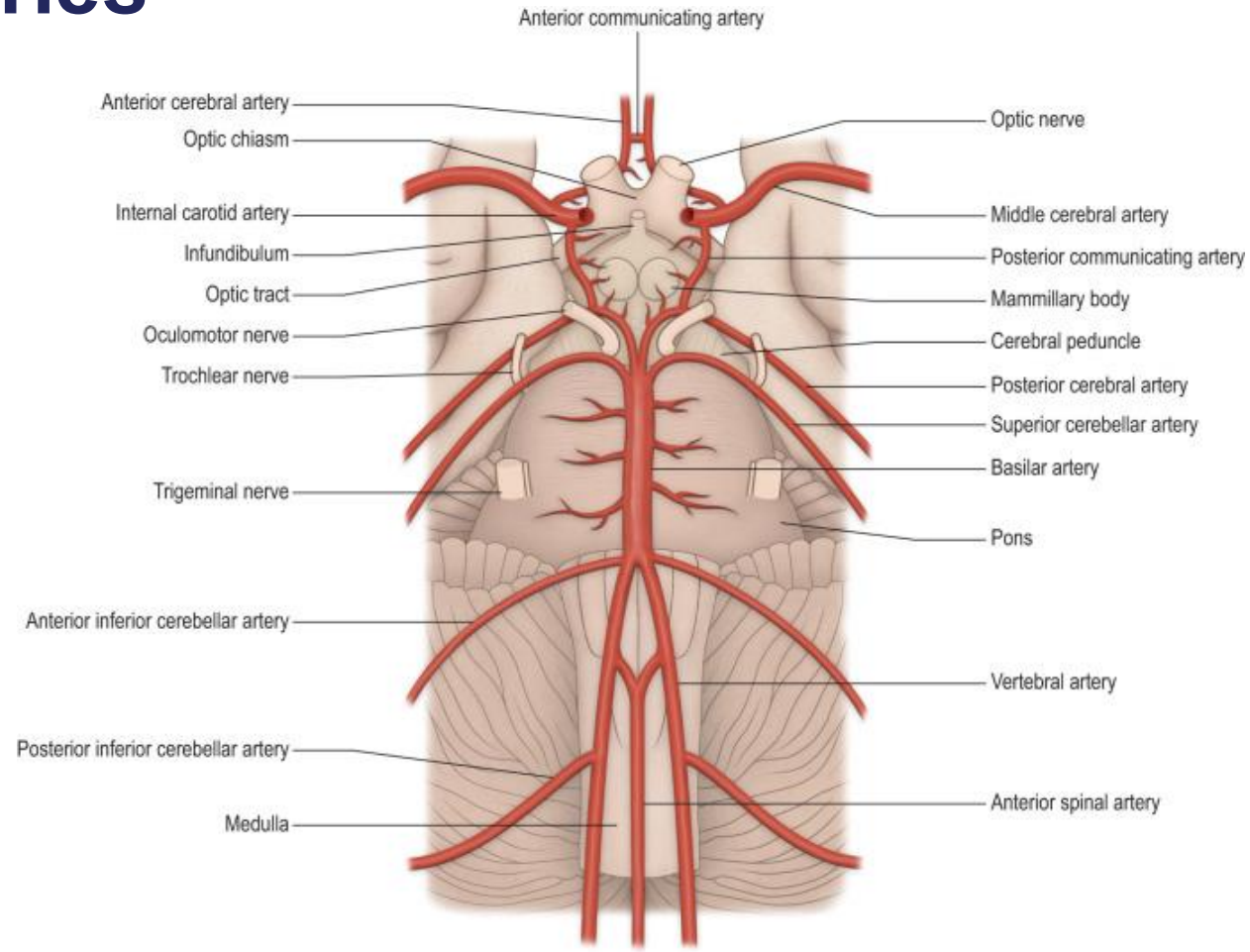


Fig. 7.2 The arrangement of arterial vessels on the base of the brain. The diagram shows the circulus arteriosus (circle of Willis).
Crossman, Neuroanatomy: An Illustrated Color Text, 6e.

Learn the anatomy now → localize vascular syndromes in Brainstem II

Brainstem and Cranial Nerves – I

Aftab Merchant, MBBS, FCPS, MHPE
Professor of Medical Education
Assistant Dean of Anatomical Education
Course Director Anatomy and Neuroscience
Tel: (702) 545-2050 | aftab.merchant@unlv.edu
Schedule an Appointment

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV